Transition History for Single-BS 5G NR CSI Localization: Model-Agnostic Gains and Their Hardware Dependence

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Abstract

A single serving base station fixes a device's range from received signal strength but not its bearing. We ask whether a short window of past measurements, the transition history, narrows this distance-ring ambiguity, in a simulated 5G NR deployment that compares a small controlled grid with a $96\,\mathrm{m}$ macro cell, and across seven estimators from k-nearest neighbors to recurrent, convolutional and attention networks. For devices that report an angle of arrival, five steps of history ($20\,\mathrm{m}$ of travel) lower the position error by 2.8 to $4.1\,\mathrm{m}$ (15 to 21%) in all seven families, and model choice barely matters. In the macro cell the gain comes entirely from the angle channels, and it is the averaging of imperfect angle reports: an unordered mean of the window reproduces it, elevation carries most of it, and with exact angles it disappears. It therefore depends on the angle error being independent between reports, falling from about 17% to 2 to 6% when the error persists. Devices without an angle gain no more than about 1% on uniform non-line-of-sight propagation, where their models beat a constant guess of the cell centre by only 5 to 6%; with mixed LOS and NLOS propagation, tree ensembles gain about $1.2\,\mathrm{m}$, an effect linked to the propagation state. Line-of-sight propagation, interfering base stations and a blocked line of sight do not change the multi-antenna gain. On the controlled grid, history raises accuracy under all three propagation settings tested, with and without an angle; there a recurrent network and a convolutional network keep improving up to fifteen steps while tree ensembles stop at about three, and shuffling the older reports inside the window, with the current report kept in place, raises the recurrent network's error at fifteen steps by 19 to 48%, so on the grid the order of the reports carries information beyond their mean.

I. Introduction

A. Motivation

Fifth- and sixth-generation cellular networks make several channel-derived measurements available natively at the base station (BS): received signal strength (RSS), signal-to-interference-plus-noise ratio (SINR), angle of arrival (AoA), and timing advance, none of which require dedicated positioning infrastructure to collect. Turning these measurements into a position estimate for the user equipment (UE) is motivated by applications ranging from emergency response and asset tracking to network-assisted navigation, and the need is sharpest precisely where satellite positioning is weakest: urban canyons, tunnels, and indoor spaces, where signal blockage and multipath substantially degrade the accuracy of the Global Positioning System (GPS). Cellular positioning targets this gap directly, since a device already in radio contact with a BS can in principle be localized from measurements the network is collecting regardless of whether positioning was requested, with no added receiver and no sensor on the device itself.

The obstacle is accuracy rather than availability. RSS, the most universally reported of these metrics, is set primarily by distance from the BS, with multipath structure and shadow fading contributing only a secondary correction, which on its own leaves position under-constrained.

Multi-BS geometric positioning resolves the missing bearing directly, but only given multiple synchronized anchors or a high-quality angle estimate. A single serving BS has neither by default. That is what makes it the harder deployment case, and the more consequential one: it is exactly the setting in which a network already capable of collecting these measurements would most benefit from a fix that adds no new infrastructure.

B. The Distance-Ring Ambiguity and the Hypothesis

Let $\mathbf{p}_{UE}$ and $\mathbf{p}_{BS}$ denote the UE and serving-BS positions, and let $r = \|\mathbf{p}_{UE} - \mathbf{p}_{BS}\|$ be their range. Under a log-distance path-loss model,

$$\mathrm{RSS} \approx P_{tx} - 10\gamma \log_{10} r + \chi, \tag{1}$$

with γ the path-loss exponent and χ zero-mean shadow fading of standard deviation σ. The map from RSS to position is therefore one-to-many: a static measurement identifies r, up to χ, but leaves bearing entirely unconstrained. Locations on a common range ring around the BS are consequently indistinguishable to a static, single-snapshot estimator, which is the *distance-ring ambiguity*. The ambiguity is worst in geometries dominated by line-of-sight (LOS) propagation, where the range-to-RSS map is close to monotone and multipath diversity is minimal. SINR and AoA can each supply the missing bearing information, but both are conditional: SINR depends on interferer placement and calibration, and AoA depends on UE beamforming hardware that a large share of connected devices does not have.

The hypothesis evaluated in this report is that a moving UE's measurement sequence encodes more positional information than any single snapshot, and that this holds regardless of the estimator applied on top of it. Two positions on the same range ring produce the same instantaneous RSS but different trajectories through measurement space, because the local channel neighborhood around each position is different and evolves differently as the UE moves.

Formally, let $\mathbf{m}(t)$ denote the vector of channel-derived measurements reported for the UE at discrete snapshot time t (Table I). Given a history depth h, the model input is the window of $h + 1$ consecutive observations,

$$\mathcal{W}_h(t) = [\, \mathbf{m}(t-h),\ \ldots,\ \mathbf{m}(t) \,], \tag{2}$$

where $h = 0$ recovers the static, memoryless fingerprint as a special case. For $h > 0$, $\mathcal{W}_h(t)$ makes the temporal derivatives of $\mathbf{m}$ observable; Section III sets out what those derivatives can and cannot constrain.

Table I: Nomenclature

Symbol	Description
$\mathbf{m}(t)$	Measurement vector at snapshot time t
h	History depth (number of prior snapshots in the window)
$\mathcal{W}_h(t)$	History window, $[\mathbf{m}(t-h), \ldots, \mathbf{m}(t)]$
$\mathbf{p}_{UE}$, $\mathbf{p}_{BS}$	UE and serving-BS positions
r	UE–BS range, $r = \lVert \mathbf{p}_{UE} - \mathbf{p}_{BS} \rVert$
θ_{az}, θ_{el}	AoA azimuth and elevation
γ	Path-loss exponent
χ	Zero-mean shadow-fading term, standard deviation σ
σ_r	Range uncertainty implied by RSS alone
N_{ant}	UE antenna count
Δt	Time between consecutive snapshots of one UE

II. Related Work

Classical RSS fingerprinting matches a measurement snapshot against a surveyed database [**?**]; single-site multipath fingerprinting establishes that one BS already observes enough structure to localize without triangulation [**?**], the principle this work extends to a moving user. Fingerprinting's characteristic failure mode is poor discrimination between points at equal range from the BS, which is exactly the distance-ring ambiguity the transition-history mechanism targets (Section I-B). The static, $h = 0$ k-nearest-neighbor regressor used as a baseline throughout this report stands in for this family.

Geometric methods recover position from an explicit system built on time-of-arrival, time-difference-of-arrival, or AoA estimates. These methods require either multiple synchronized BSs or a high-quality angle estimate, so the single-serving-BS, single-angle-source deployment studied here is precisely the regime in which the geometric system is underdetermined: for a device that reports no angle at all, the available geometry constrains range but leaves bearing unresolved.

Recurrent and dynamic-system formulations have likewise been applied to channel state information (CSI)-based tracking [**?**], [**?**], [**?**], and a related line shows that the temporal autocorrelation of RSS improves fingerprinting accuracy when the signal is treated as a stationary process observed at a fixed location [**?**], [**?**]. That precursor is complementary rather than nested: its temporal signal is order-invariant and is aggregated across access points, so it survives for a stationary receiver and vanishes for a moving one, whereas the transition-history window used here is ordered, is read from a single BS, and encodes exactly the motion the in-place statistic discards. More broadly, this body of work motivates a new architecture and reports that it lowers error; the framing adopted here instead holds the estimator family fixed and varies only the input window, isolating the effect of the input transformation from the choice of architecture and allowing the comparison to extend by design to model families not tested in this report.

III. System Model and Problem Formulation

A. Measurement Model

At discrete snapshot time t the serving BS observes

$$\mathbf{m}(t) = \big[\, \mathrm{RSS}(t),\ \mathrm{SINR}(t),\ \theta_{az}(t),\ \theta_{el}(t) \,\big], \tag{3}$$

with the azimuth/elevation pair $(\theta_{az}, \theta_{el})$ present only for UEs whose antenna count $N_{ant} \geq 2$. Multi-antenna UEs are modeled by maximum-ratio combining over the per-element channel,

$$H_{\text{eff}}(f) = \sqrt{\sum_{i=1}^{N_{ant}} |H_i(f)|^2}, \tag{4}$$

with a device-specific antenna-gain offset applied additively in dB. Throughout this report, UEs with $N_{ant} \geq 2$ are referred to as multi-antenna devices and UEs with $N_{ant} = 1$ as single-antenna devices.

B. What History Can Add

Over the window $\mathcal{W}_h(t)$ of Equation (2), an estimator can form finite differences of the measurements, of which two carry geometric meaning. The rate of change of RSS,

$$\frac{d\,\text{RSS}}{dt} \approx -\frac{10\gamma}{\ln 10} \cdot \frac{\dot{r}}{r}, \tag{5}$$

reflects the component of motion along the BS bearing, and the angular rate,

$$\dot{\theta}_{az} = \frac{v_\perp}{r}, \tag{6}$$

where AoA is available, gives the tangential component of the velocity $v_\perp$ and, jointly with $\dot{r}$, the heading. For a UE moving at roughly constant speed, the pair $(\dot{r}, \dot{\theta}_{az})$ constrains position to the intersection of a range ring and a motion-consistent arc, a far smaller set than the ring alone. This geometric reading motivates the hypothesis; Section IX tests it against a simpler statistical one, in which a window helps only because it offers several noisy views of a nearly unchanged angle.

Without AoA, only the first of these is available, and its usefulness is bounded by the path-loss gradient. Differentiating Equation (1) with respect to range gives

$$\left| \frac{d\,\text{RSS}}{dr} \right| = \frac{10\gamma}{r \ln 10} \text{ dB/m}, \tag{7}$$

so a step of fixed length moves RSS by an amount that falls as $1/r$, while the shadow-fading term χ does not shrink with range. A sequence of RSS values can therefore carry useful information about motion relative to the BS only where the per-step change is not swamped by σ; how far from the BS that condition holds is an empirical question that the two study scales in this report bracket from either side.

C. Task Formulations

Two task formulations are evaluated. Classification predicts one of G discrete grid cells, and is appropriate when grid spacing greatly exceeds position jitter. Regression predicts a continuous position relative to the serving BS and is scored by the 2D Euclidean error in meters. Regression is used for the larger deployment, where a 625-class classifier proved memory-prohibitive and where evaluation is over unseen users rather than unseen time steps of known users.

IV. Experimental Setup

A. Channel Simulation

All data is generated with a 3GPP TR 38.901-compliant channel simulator at $100\,\text{MHz}$ bandwidth (the early single-user grid sweeps used $20\,\text{MHz}$), 256 orthogonal frequency-division multiplexing (OFDM) subcarriers, $30\,\text{dBm}$ transmit power per site, and an omnidirectional BS antenna element. The simulator's spatial-consistency model is the physical precondition for the mechanism under test: the channel evolves continuously along a UE's trajectory rather than being redrawn independently at each step, so consecutive snapshots carry correlated, motion-dependent structure. This work extends a time-aware multi-user massive multiple-input multiple-output (MIMO) simulation infrastructure developed within the same research group, adding grid-constrained trajectory generation and the device-heterogeneity model of Section IV-D.

Where a deployment includes interfering BSs in addition to the serving BS, SINR is computed as

$$\text{SINR} = \frac{P_{\text{serving}}}{\sum_i P_{\text{IBS-}i} + P_{\text{noise}}}, \qquad P_{\text{noise}} = -104\,\text{dBm}, \tag{8}$$

so interference is contributed only by the modeled interfering sites. No user-to-user coupling is represented: users simulated in a shared batch share large-scale channel parameters, but a user's SINR does not depend on how many other users are simulated alongside it.

B. Studies

The evidence comes from three simulated deployments at two scales, summarized in Table II. The *controlled-grid study* places UEs on a 15×15 grid at $2\,\text{m}$ pitch within $20\,\text{m}$ of the serving BS and predicts the exact cell, under conditions chosen to be clean and repeatable; a single-user variant of the same set-up sweeps the grid from 3×3 to 20×20 cells. The *macro-cell study* places 300 users on a 25×25 grid at $4\,\text{m}$ pitch, 21 to $157\,\text{m}$ from an off-grid serving BS, predicts continuous position, and evaluates on users never seen during training. The 25×25 *regression runs* share the macro-cell geometry, simulate five device profiles (one user each) pooled in one model, split in time, and are repeated under three propagation settings (UMi line of sight, uniform NLOS, per-cell mixed); they are used here only for the case of RSS and SINR without AoA. A fixed history depth of $h = 3$ is used in the controlled-grid study and $h = 5$ in the macro-cell study, with sweeps over h where stated.

In the controlled-grid study the serving BS sits either at the grid center, at $[19, 19, 10]\,\text{m}$, where the grid spans the full $360°$ of azimuth, or beyond the north-east corner, at $[48, 48, 10]\,\text{m}$, where all 225 cells fall within a wedge of about $52°$. Two interfering sites, at $[-11, 19, 10]$ and $[19, -11, 10]\,\text{m}$, enter through Equation (8) for both placements. All walks run the same urban-micro NLOS scenario at $3.5\,\text{GHz}$; the four Voronoi zones are position labels for per-zone breakdowns and differ in no propagation property. Figure 1 draws both deployments at the same scale.

In the macro-cell study the serving BS sits off-grid to the north-east at $[116, 116, 10]\,\text{m}$, with two interfering sites at $[-60, 53, 10]$ and $[53, -60, 10]\,\text{m}$ entering through Equation (8). The primary realization of this study uses a single urban-micro non-line-of-sight (NLOS) propagation profile for every user. A second realization assigns each of four Voronoi zones its own LOS or NLOS profile and generates each scenario segment of a trajectory as an independent channel; it serves to test how propagation conditions change the result.

A fourth, targeted deployment, the *blocked line-of-sight scenario*, follows UEs walking a straight $200\,\text{m}$ corridor whose middle section passes behind an obstruction, so that the link

Table II: The three simulated deployments. The macro-cell study is reported per device group throughout.

	Controlled-grid study	25×25 regression runs	Macro-cell study
Task	Classification (225 cells)	Regression	Regression
Grid, pitch	15×15, 2 m	25×25, 4 m	25×25, 4 m
Service area	28×28 m	96×96 m	96×96 m
Range to serving BS	0–20 m (center BS); 21–61 m (corner BS)	21–157 m	21–157 m, mean 93 m
Carrier	3.0 GHz	3.0 GHz	3.0 GHz
Population	5 device profiles, 450,005 samples	5 device profiles, one user each, 1,250,000 samples	300 users simulated, 232 retained, 47 tested per split
Split	Chronological $80/20$ within each user	Chronological, or shuffled 5-step blocks	$80/20$ over disjoint users
Mobility	Random walk, 1.5 m/s	Random walk	Four patterns, 0.1–15 m/s
AoA impairment	SINR-dependent noise, 5° quantization	4° Gaussian noise, 5° quantization (AoA not used in the runs cited here)	4° Gaussian noise, 5° quantization
Repetitions	Single run	Single run	Five seeds for headline comparisons

goes from LOS to NLOS and back. The width of the blocked strip is varied from 5 to 150 m and UE speed takes values of 1.5, 4 and 10 m/s. The BS stands 500 m from the corridor so that range is nearly constant along it, which keeps the change in propagation from being confounded with a change in distance.

C. Mobility

The controlled-grid study moves each UE on a random walk over the 8-connected grid graph at 1.5 m/s; the walk carries no persistent heading. In the macro-cell study each user follows one of four trajectory patterns: straight-line navigation that reflects off the area boundary (35% of users), a random walk that continues straight with probability 0.7 (35%), a tour between five sampled waypoints (20%), or a straight transit across the area (10%). Each user also belongs to one of four speed classes: pedestrian, 1.0–1.5 m/s (40%); jogger, 2.5–4.5 m/s (25%); vehicle, 8–15 m/s (25%); or near-static, 0.1–0.5 m/s (10%). All four patterns carry heading persistence, so the macro-cell mobility is favorable to history rather than adversarial to it.

In both studies every step of a walk advances exactly one grid cell, and the time between snapshots is the step length divided by the UE's speed. A fixed history depth is therefore a fixed distance travelled, h grid steps, but not a fixed duration: across the 232 retained macro-cell users the time between snapshots Δt ranges from 0.27 to 32.5 s, with a median of 2.71 s. Position jitter of ± 0.1 m is added to every snapshot so that no estimator can latch onto exactly repeated coordinates.

D. Device Model and AoA Impairment

The controlled-grid study simulates five device profiles: two four-antenna devices that differ only in their walk, a two-antenna device, a one-antenna device, and a two-antenna device carried at 0.9 m height. The macro-cell study draws from a 3GPP-representative population of 85% multi-antenna devices (about 45% with four antennas and 40% with two) and 15% single-antenna devices, each with its own antenna gain and carrying height. Antenna count enters the channel

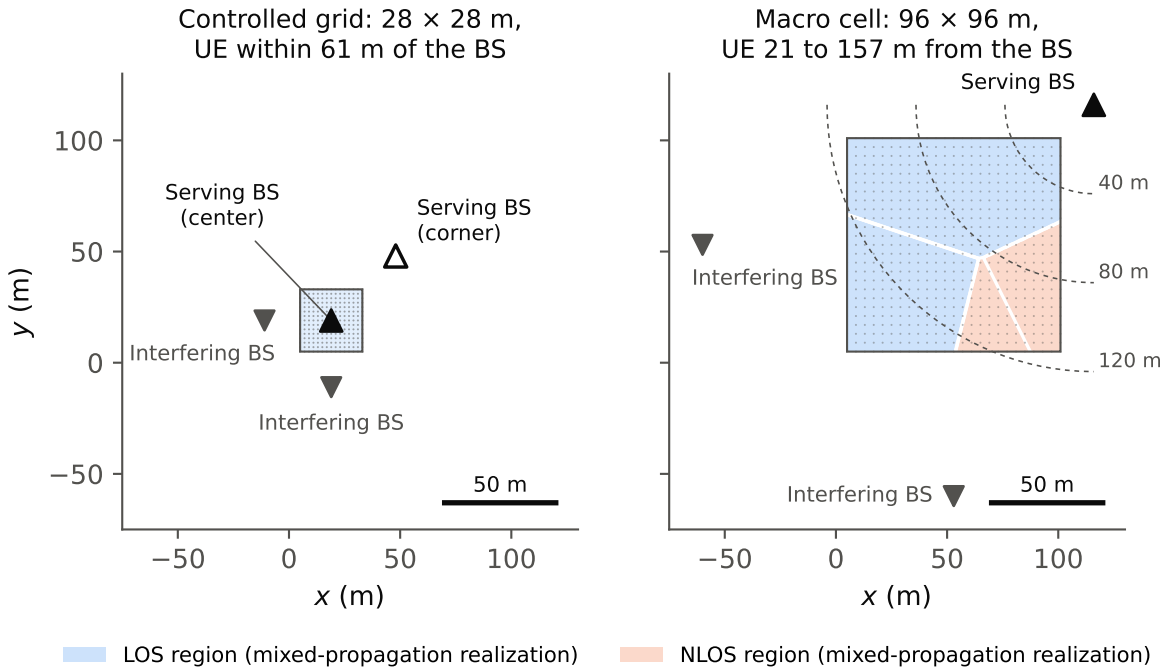


Figure 1: The two deployments drawn at the same scale, with identical axes and a $50\,\mathrm{m}$ scale bar in each panel. Left: the controlled grid, 15×15 cells at $2\,\mathrm{m}$ pitch, with its two serving-BS placements (filled: center; open: corner) and two interfering BSs. Right: the macro cell, 25×25 cells at $4\,\mathrm{m}$ pitch, with the serving BS beyond its north-east corner, two interfering BSs, and range from the serving BS at 40, 80 and $120\,\mathrm{m}$. Shading marks the macro cell's four Voronoi regions, park to the north and highway to the south-west (LOS), shopping district to the south-east and residential area to the south (NLOS), with the propagation each receives in the mixed-propagation realization; in the primary realization every region shares the same NLOS profile.

through Equation (4); the AoA capability of a device is then imposed when features are built, so the two device groups share an identical channel model and differ only in whether an angle is reported.

Clean simulated arrival angles are degraded before use. In the controlled-grid study the noise on each axis depends on SINR,

$$\sigma_{\mathrm{AoA}}(t) = \mathrm{clamp}\Big(2.0^\circ \cdot 10^{-\frac{\mathrm{SINR}_{dB}(t)-10}{15}},\ 1.0^\circ,\ 20.0^\circ\Big), \tag{9}$$

while the macro-cell study applies a fixed 4° Gaussian error, drawn independently for every report, an assumption that Section IX-A shows to matter. Both then quantize azimuth and elevation to a 5° codebook. Single-antenna devices cannot form a beam and report no angle; in the macro-cell features their angle is set to 0°.

E. Features

In the controlled-grid study, $\mathbf{m}(t)$ contains RSS and SINR, optionally extended with the AoA pair and with three device parameters: antenna count, antenna gain, and carrying height. The window of Equation (2) is built per user after sorting by time; the first h snapshots of each user are dropped, and no window crosses from one user to another. Values are stacked in absolute

form, because device-specific RSS and SINR offsets are stable across the grid and themselves act as fingerprints.

In the macro-cell study, each snapshot carries 13 channels: RSS, SINR, the sines and cosines of both angles, which remove the $\pm 180^\circ$ wrap discontinuity, two ray-projection coordinates formed from a path-loss range estimate and the AoA unit vector, the time increment Δt, and first differences of RSS, azimuth, and the two ray coordinates with respect to the previous snapshot, each divided by Δt. Every derived channel is computable from the raw measurements, so the derived set adds inductive bias rather than information. Because the first differences are part of each snapshot, the $h = 0$ input of the macro-cell study already contains one step of change; it is the current snapshot together with its difference from the previous one, and the $h = 5$ input stacks six such snapshots. Three static device features, antenna count, antenna gain, and carrying height, accompany every input.

F. Estimators

The macro-cell study evaluates seven estimators spanning five algorithmic paradigms, listed in Table III: k-nearest neighbors (k-NN), Random Forest, XGBoost, a gated recurrent unit (GRU) network, and three one-dimensional convolutional neural networks (1D-CNN), one with an AoA-validity channel and one with temporal self-attention. Hyperparameters are held fixed across history depths, so the input window is the only factor that varies within a model. The four neural models share a training recipe, AdamW at a learning rate of 3×10^{-4} with weight decay 10^{-2}, batch size 512, a Huber loss on position with auxiliary speed and uncertainty heads, and at most 60 epochs with early stopping. The controlled-grid study uses Random Forest and XGBoost classifiers, and its grid-size sweep adds a Gaussian likelihood model and a multilayer perceptron.

Table III: Estimators used in the macro-cell study.

Paradigm	Model	Configuration
Instance-based	k-NN	$k = 5$
Bagged trees	Random Forest	150 trees, depth 16, min. leaf 5
Boosted trees	XGBoost	50 trees, depth 5, learning rate 0.15
Recurrent	GRU	2 layers, 254 K parameters
Convolutional	1D-CNN	3 convolutional layers, 67 K parameters
Convolutional	Mask-aware 1D-CNN	1D-CNN with an AoA-validity channel, 67 K parameters
Attention	1D-CNN with temporal self-attention	200 K parameters

G. Evaluation Protocol

The controlled-grid study splits each user's walk in time, training on the first 80% of steps and testing on the last 20%; a random split would leak, because spatial consistency makes temporally adjacent samples near-duplicates. It is scored by exact-cell accuracy. The macro-cell study splits over users, so that the 47 test users of each split are never seen during training, and it is scored by the 2D Euclidean error between predicted and true position: its mean, median, and 90th percentile, and the fraction of predictions within 5, 10 and 20 m. Every macro-cell result is reported separately for multi-antenna and single-antenna devices.

The primary comparison is the change in error when history is added, measured within one model against its own $h = 0$ input. For the macro-cell headline results the whole protocol is repeated over five seeds, each of which draws a different user split and model initialization. Two uncertainties follow. The marginal spread across seeds measures how much a single reported

number could move by chance, and it is dominated by which users happen to land in the test set. The seed-paired difference compares two inputs on the same split and so cancels that source; it is the appropriate test for a change of input, and its 95% confidence interval is reported with each gain.

The neural models select their stopping epoch by error on the test users. This applies equally to both depths of a model, so within-model comparisons are unaffected, but it makes the absolute errors of the neural models, and any comparison between a neural model and a tree or instance-based model, optimistic in the neural model's favor.

V. Results: The Controlled Grid

The controlled grid was simulated under three propagation settings, each with the BS at the grid center and beyond its corner: RMa line of sight for every walk, a mixed setting in which each Voronoi cell has its own UMi line-of-sight or non-line-of-sight scenario, and uniform UMi non-line-of-sight. The mixed setting is the reference set of this section, because it is the only one in which the propagation changes along a walk; the other two bracket it. Without AoA, adding three steps of history to RSS and SINR raised exact-cell accuracy in all twelve configurations (three propagation settings, two placements, two classifiers; Figure 2). XGBoost gained between 13.2 and 27.4 percentage points, which is 2.6 to 4.3 m less mean error, and Random Forest between 4.9 and 13.0 points, or 1.0 to 3.1 m. A fourth set, uniform NLOS at 3.5 GHz, agrees with the 3.0 GHz one within one to two points, so the carrier does not matter here. The absolute accuracy, in contrast, moves by tens of points with the propagation: with the BS at the center, XGBoost with history reaches 29.6% under uniform NLOS, 50.6% under RMa line of sight and 59.1% under the mixed setting. The snapshot models are weak and differ between classifiers (mixed setting, center BS: 31.7% for XGBoost and 53.5% for Random Forest), so part of XGBoost's larger gain is recovery from a poor snapshot model; the gains of both classifiers are positive in every set. Always guessing the center of the grid gives a mean error of 10.9 m on the test samples.

The placement change that leaves history's contribution positive removes most of AoA's. With history already in the input, adding AoA raised XGBoost's accuracy by 20.9, 19.5 and 38.5 points at the center BS in the RMa, mixed and NLOS settings, but by only 4.0, 1.0 and 6.2 points at the corner, and Random Forest's by 13.7, 9.9 and 31.9 against 2.8, -0.4 and 5.8 points. From the corner the whole grid falls within a wedge of about 52°, and once a 5° codebook and a few degrees of estimation noise are applied, azimuth no longer separates neighboring cells. The remaining gain at the corner is consistent with elevation, which encodes distance, but this grid study does not isolate it. With AoA available, history adds less at the center BS (0.2 to 0.6 m) than at the corner (0.7 to 2.0 m), where the angle is weaker and leaves more error to average away. This loss of AoA's value is specific to a short-range grid with 2 m cells; it is not a general property of corner placements.

Where history was swept one step at a time, most of the gain arrived with the first step (Figure 3). On a single-user 7×7 grid, one step of history raised accuracy by 10 to 12 points for each of XGBoost, Random Forest and a multilayer perceptron, and the next two steps added about 5 to 7 points between them. With AoA available, the same profile appears in a uniform urban-macro NLOS scenario, where accuracy rose from 29% to 46% at the first step and to 51% by $h = 3$. In the corresponding LOS scenario, where AoA alone already placed 93% of samples in the right cell, history added 3 points at the first step and nothing after it.

The direction of the history effect is also stable across scale and estimator (Figure 4). In the single-user version of the study, history raised accuracy at every grid size from 3×3 to 20×20

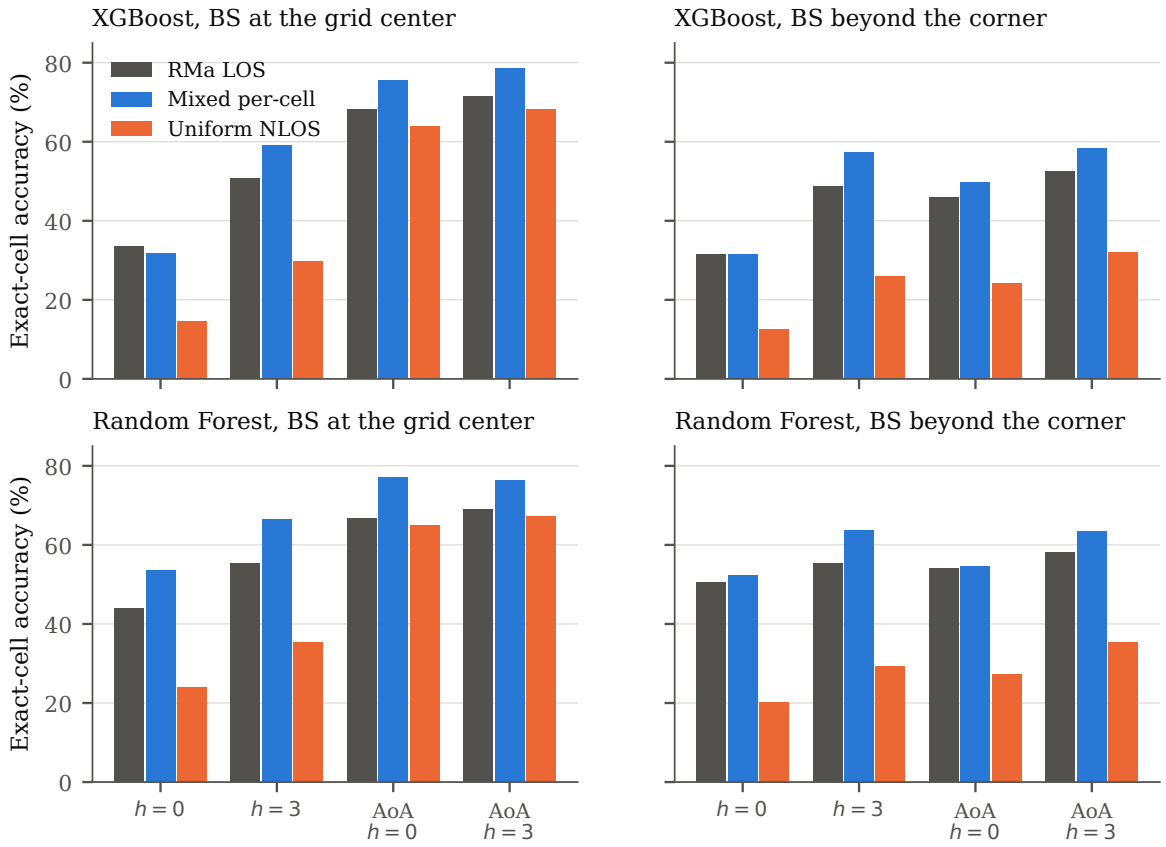


Figure 2: Controlled grid: exact-cell accuracy from RSS and SINR, and with AoA added, at history depth $h = 0$ and $h = 3$, for XGBoost (top) and Random Forest (bottom), with the BS at the grid center (left) and beyond its corner (right), under three propagation settings. Single run per bar, 3.0 GHz.

cells, by between 9.2 and 21.8 points, and on the 7×7 grid it raised the accuracy of all four estimators tried, from 4.7 points for a Gaussian likelihood model to 18.2 points for XGBoost.

Longer windows separate the model families (Figure 5). The tree ensembles take their gain at one to three steps and then stop: XGBoost is flat from $h = 3$ to $h = 15$ (mixed setting, corner BS, RSS and SINR: 3.58 m to 3.46 m), and Random Forest is best at three to five steps and worse at fifteen (2.78 m to 3.35 m). A one-dimensional CNN and a GRU keep improving with every added depth: the CNN goes from 8.34 m at $h = 0$ to 2.49 m at $h = 15$ and the GRU to 1.42 m, against 3.35 and 3.46 m for the trees. With AoA the ordering is the same, with the GRU at 1.30 m against 2.82 to 3.15 m for the trees; with the BS at the center the GRU reaches 0.60 m against 0.99 m for XGBoost. The same ordering holds in all four sets tested. History therefore helps every family, but the neural models use a longer window than the trees can, and the final ranking of the families at long history is the reverse of the ranking at no history. An LSTM of the same size, tried on the mixed setting with the BS beyond the corner, performs the same as the GRU (1.40 against 1.42 m at $h = 15$ without AoA), so the gain comes from using the sequence rather than from the particular recurrent cell.

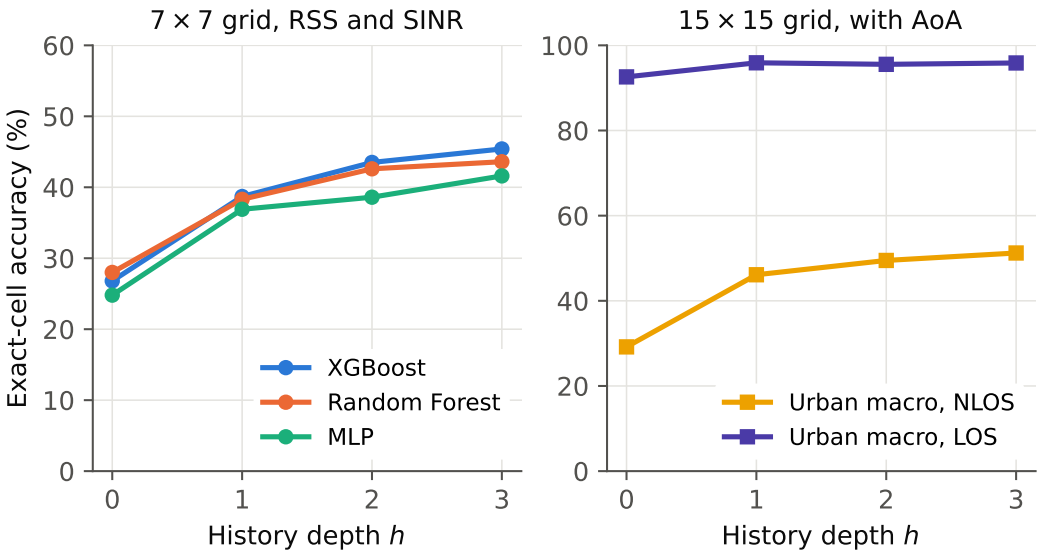


Figure 3: Controlled grid, single user: exact-cell accuracy as history depth grows. Left: 7×7 grid with RSS and SINR, three classifiers. Right: 15×15 grid with AoA in a single urban-macro propagation scenario, NLOS and LOS. Single run per point.

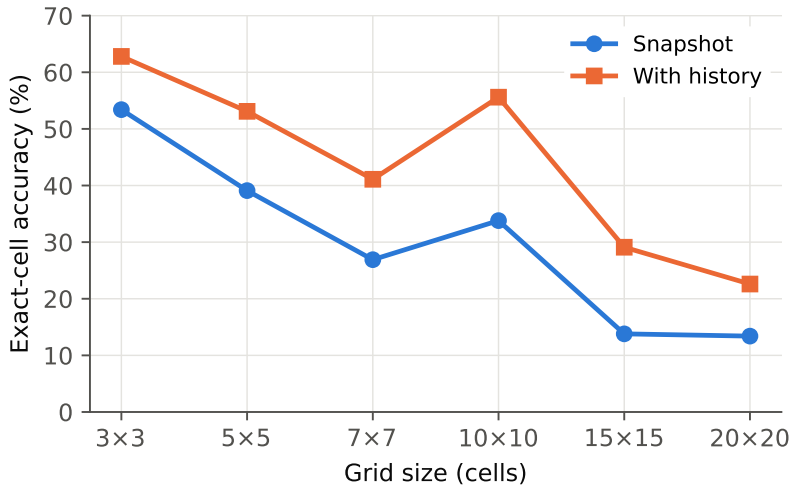


Figure 4: Controlled grid, single user: exact-cell accuracy with and without history as the grid grows from 3×3 to 20×20 cells, XGBoost with RSS and SINR at the best depth up to $h = 3$. Single run per grid size.

Two controls point to what the neural models use. Giving the trees only the current report and the mean of the window, which is what an averaging account of history would predict to be enough, does not make them keep improving: without AoA they are worse than the trees on raw lags from $h = 3$ on (mixed setting, corner BS, $h = 5$: XGBoost 4.52 m against 3.51 m), and with AoA they only equal them. And permuting the older reports inside each window, in training and in test, while the current report keeps its place, raises the error of the GRU at $h = 15$ by 19 to 48% in all eight combinations of set, placement and angle: without AoA the mixed-setting GRU

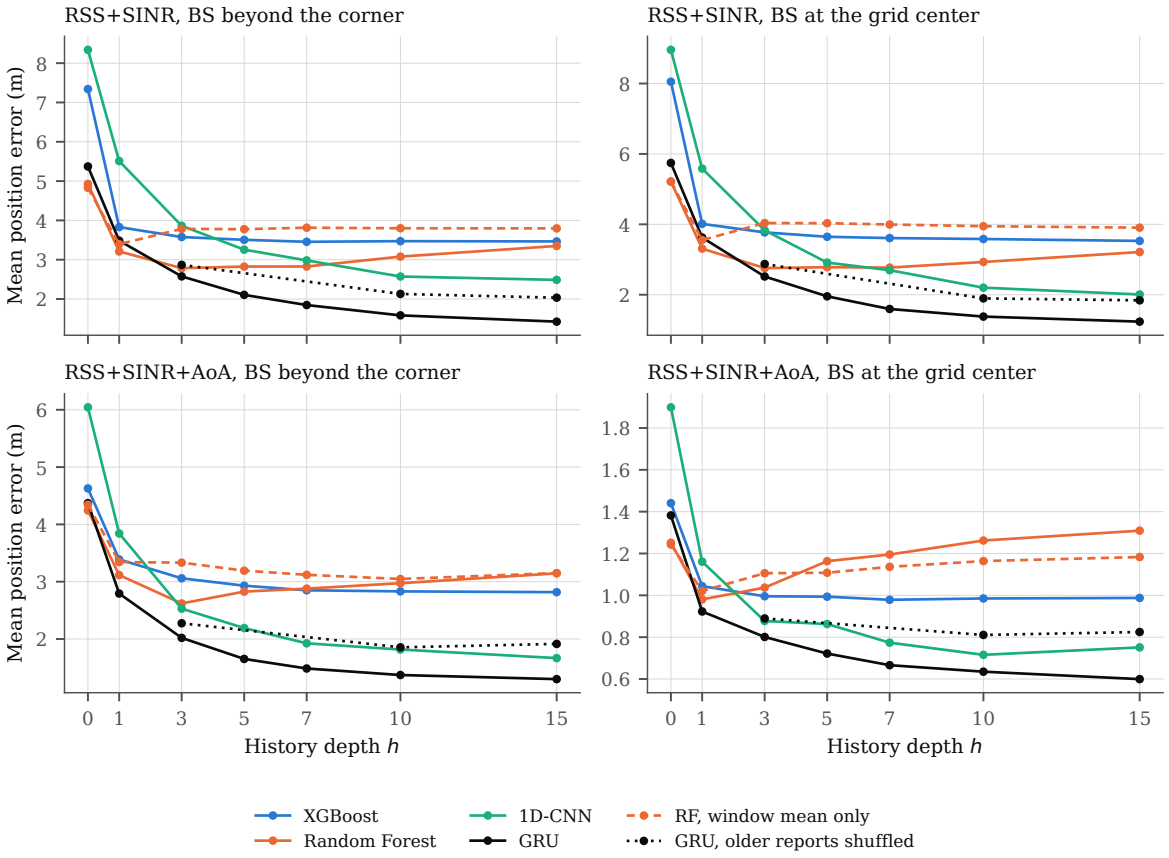


Figure 5: Controlled grid, mixed propagation: mean position error against history depth h without AoA (top) and with AoA (bottom), BS beyond the corner (left) and at the center (right). Tree models use raw lags (solid) or only the current report and the window mean (dashed); the GRU is also shown with the older reports shuffled inside each window and the current report kept in place (dotted). Trees and window-mean trees: single run; CNN and GRU: three seeds; shuffled GRU: two seeds.

with the BS beyond the corner goes from 1.42 to 2.03 m, and the shuffled network gains little beyond ten steps while the ordered one keeps gaining. The shuffled GRU is still better than the best tree in all eight combinations, if only narrowly in some, so only part of its advantage depends on order. The most economical reading is that, on this grid, a single RSS and SINR report is ambiguous between cells that lie on the same signal contour around the BS, that the mean of a window does not remove that ambiguity, and that the ordered sequence of reports, which traces the movement across the contours, does. The history gain on the grid would then have two parts: an averaging part, shared by all models and the only part isolated in the macro-cell study, and a sequence part that only recurrent and convolutional models exploit. This is an interpretation supported by four sets and two seeds in the shuffled test, not a direct proof. A first version of the test, which also moved the current report, collapsed the networks below the accuracy of a snapshot model; that showed only that the current report must be identifiable, and it is not used.

Because RSS and SINR are stacked in absolute form, part of what a history window carries is a stable, device-specific offset, and history could in principle be helping only by identifying the

device. Supplying the device parameters directly rules this out as the whole story. In the mixed setting with the BS at the center, the parameters alone raise XGBoost from 31.7% to 59.2%, as much as history does (59.1%), and the user identifier alone is as informative (61.8%); but history still adds on top of them, 7.5 points for XGBoost and 6.1 for Random Forest (59.2% to 66.7% and 65.7% to 71.8%). With AoA available, history adds less (2.2 points for XGBoost on top of the device parameters).

Every number in this section comes from a single run unless stated, and the same configuration evaluated in two separate runs differed by 2 to 3 points in an earlier version of this study, so only differences well above that are read as effects. The depth and scale sweeps above come from earlier single-user simulations that were not regenerated. The controlled grid also tests later time steps of users seen in training, at no more than 61 m from the BS. Whether history keeps its value at macro-cell range, for users the model has never seen and a population with mixed hardware, is the question the macro-cell study answers.

VI. Results: What Accuracy a Macro Cell Allows

At the distances of the macro-cell study, RSS alone pins down range only to within tens of meters, and that sets the scale against which every error in the following sections should be read. Fitting Equation (1) to all $144{,}437$ macro-cell snapshots, with the antenna gain and antenna count of each device as controls, gives $\gamma = 3.26$ and a residual shadowing spread of $\sigma = 10.1$ dB; range explains only 20% of the variance in RSS. Dividing σ by the gradient of Equation (7) turns this into the range uncertainty σ_r implied by a single RSS reading: 40 m at the 10th percentile of macro-cell range (56 m), 68 m at the median (95 m), and 90 m at the 90th percentile (126 m). The same calculation gives 7 m at 10 m from the BS, the distance typical of the controlled grid. The two studies therefore differ not only in size but in how much a single RSS value says about distance, by roughly an order of magnitude. This is a ranging resolution and not a baseline: a model that ignored every measurement and always predicted the mean training position would be wrong by about 35 m (Table IV).

Table IV sets this physical reference beside measured errors for both device groups, and beside the error of the simplest possible predictor, which always returns the mean training position: 35.3 m for multi-antenna devices and 38.5 m for single-antenna devices. A snapshot k-NN fingerprint, the classical approach, misses multi-antenna devices by 23 m on average and single-antenna devices by 46 m, which for single-antenna devices is worse than the constant guess. The trained models do considerably better for multi-antenna devices: over 20 user splits XGBoost is 47% below the constant guess at $h = 0$ and 56% below it at $h = 5$. For single-antenna devices they do not: XGBoost is 6% below it at $h = 0$ and 5% at $h = 5$, and the two groups remain about 20 m apart in every model.

The two groups are clearly separated in the error distribution, though not disjoint (Figure 6): their error histograms overlap by 52% with history and 63% without in the plotted split. With five steps of history, XGBoost places 34% of multi-antenna predictions within 10 m of the true position and 71% within 20 m, with a median error of 13.7 m and a 90th percentile of 29.5 m. For single-antenna devices the corresponding fractions are 7% and 21%, the median is 34.7 m, and the 90th percentile is 59.1 m. Fewer than 12% of multi-antenna predictions and 2% of single-antenna predictions fall within 5 m.

The two groups also fail in different directions. Splitting each error into its component along the line to the BS, the radial or range error, and the component across it, the tangential or cross-range error, shows that multi-antenna devices are limited by range. With history, their mean radial error is 14.1 m against a tangential error of 4.6 m, so 90% of their error variance lies along the line to

Table IV: Reference error levels in the macro-cell study, per device group. Measured errors are means over five seeds, and the constant-guess errors means over 20 seeds; neural-model errors are optimistic because their stopping epoch was chosen on the test users. The ranging resolution is a standard deviation, not an error.

Reference	Multi-antenna	Single-antenna
Constant guess (mean training position)	35.3 m	38.5 m
RSS ranging resolution σ_r at median range	68 m	68 m
k-NN, $h = 0$	23.4 m	46.2 m
XGBoost, $h = 0$	18.8 m	35.8 m
XGBoost, $h = 5$	15.7 m	35.8 m
Best of seven models, $h = 5$	15.5 m	34.5 m

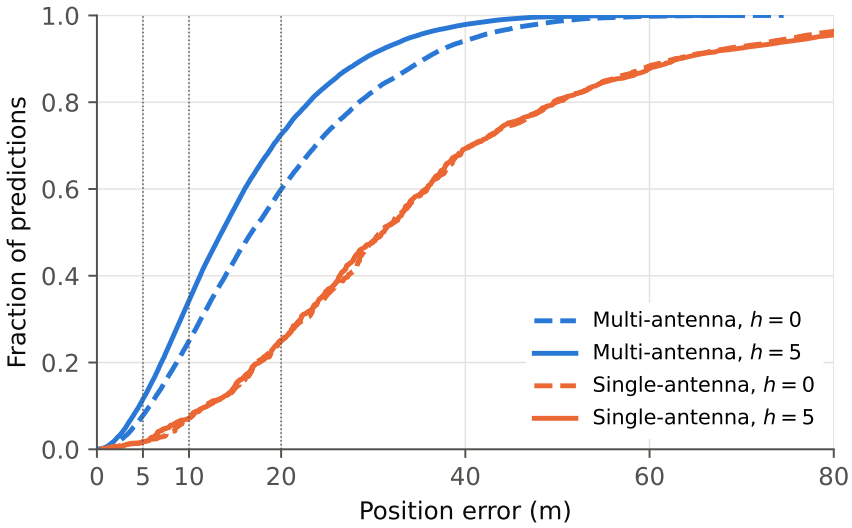


Figure 6: Distribution of position error per device group, XGBoost without history ($h = 0$, dashed) and with five steps of history ($h = 5$, solid), macro-cell study, one seed. Dotted lines mark 5, 10 and 20 m.

the BS. That radial error, 17.5 m in root-mean-square terms, is about four times smaller than the 68 m standard deviation of a single-reading range estimate, so the models extract considerably more range information than a path-loss inversion. Single-antenna devices fail differently. Their tangential error of 25.3 m exceeds their radial error of 20.1 m, and about 60% of their squared error is tangential, which is the distance-ring ambiguity of Section I-B observed. But they are not placed at the right distance: their radial error is larger than that of multi-antenna devices, the predicted range correlates only 0.51 ± 0.06 with the true range, and predictions are pulled toward the BS by about 10 m on average. Over 20 splits these devices are only 5 to 6% better than the constant guess of Table IV.

Error does not simply grow with distance from the BS (Figure 7). For multi-antenna devices the band from 80 to 120 m, which holds about half of the test samples, has the smallest error, while both the nearest band and the farthest have larger errors. The nearest band is also where history makes the largest difference, cutting the mean error from 21.3 to 11.1 m. Single-antenna

error rises from about 30 m within 80 m of the BS to about 40 m beyond 120 m, and history changes it in none of the bands. Only about 160 single-antenna test samples per split fall within 40 m of the BS, so that band is the least certain.

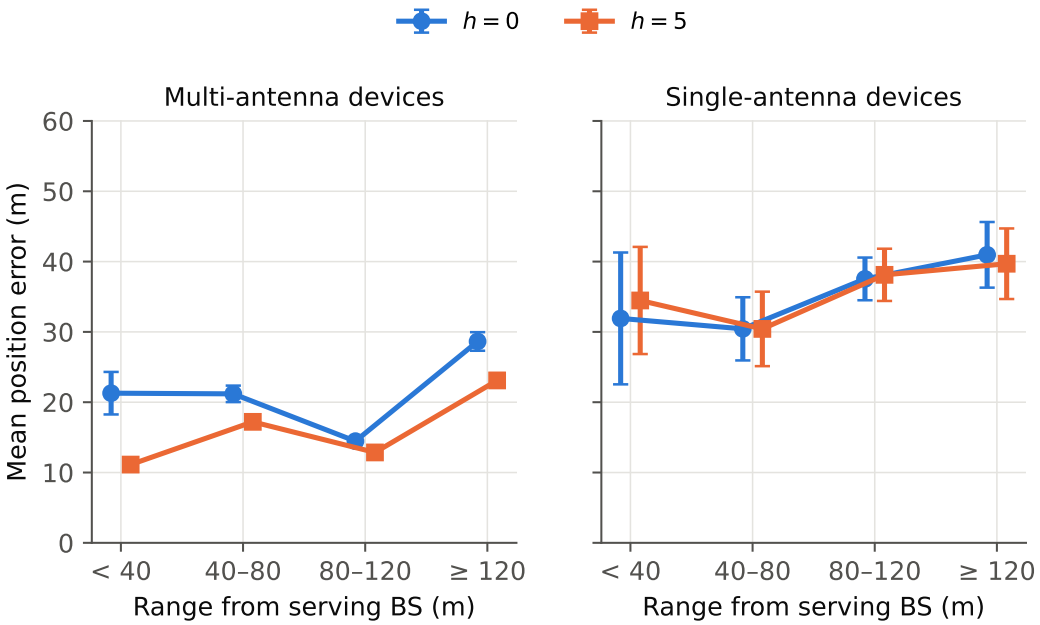


Figure 7: Mean position error by range from the serving BS, per device group, XGBoost without and with history, macro-cell study. Means over five seeds; bars are 95% confidence intervals across seeds.

The gap between the controlled grid's sub-meter errors and the macro cell's tens of meters follows from these references rather than from a failure of the models. Beyond the order-of-magnitude difference in what one RSS reading says about range, the macro cell covers twelve times the area, offers about nine times fewer training samples per grid cell, and tests on users the model has never seen. Because the two device groups fail in such different ways, the effect of history has to be measured on each of them separately.

VII. Results: Multi-Antenna Devices

For multi-antenna devices, history lowered error in every one of the seven model families, by 2.8 to 4.1 m, or 15 to 21% of each model's own $h = 0$ error (Figure 8; per-model values in Table VII). Each gain is a seed-paired difference over five independent user splits, and none of the seven confidence intervals comes close to zero. The gain is of similar size across instance-based, bagged, boosted, recurrent, convolutional and attention-based models, which is the behavior expected if history supplies information that the $h = 0$ input lacks, rather than suiting one architecture's inductive bias. These gains are measured under the angle impairment of Section IV, in which every report carries an independent 4° error; Section IX-A shows how much of the gain depends on that.

The gain builds over the first few steps of history and then levels off (Figure 9, left). With the exception of k-NN, whose error rises again between $h = 5$ and $h = 10$, and Random Forest, which is flat over that interval, every family is still improving at $h = 10$, or 40 m of travel. For XGBoost the overall error varies by only about 0.05 m between $h = 10$ and $h = 30$. In follow-up

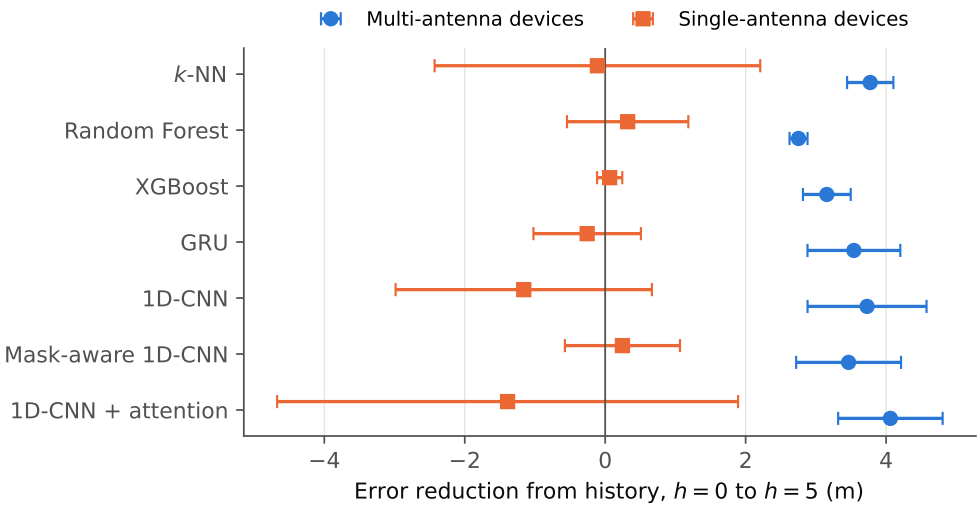


Figure 8: Reduction in mean error from adding history ($h = 0$ to $h = 5$), per model and device group, macro-cell study. Each marker is the mean seed-paired difference over five user splits; bars are 95% confidence intervals. Positive values mean history lowered the error.

runs over 10 to 20 user splits, XGBoost's gain from the full window is 1.2, 2.5, 3.2, 3.5 and 3.7 m at 1, 3, 5, 10 and 20 steps, so most of it is in place by 10 steps. No single depth is optimal across families; what distinguishes them is whether an over-long window costs anything. A tree ensemble can decline to split on an uninformative lag, whereas k-NN measures distance over the whole concatenated window, so each stale lag adds noise to its metric; this account is offered as an interpretation of the depth curves rather than as a separately tested result.

Because each step covers one grid cell, history depth measures distance travelled rather than elapsed time, and the speed classes of the population separate the two (Figure 10). Measured against path length, the four classes improve over a similar span: every class has gained most of its benefit within 12 to 20 m. Measured against time, the same curves are spread across two orders of magnitude, and the 1D-CNN's best window lasts 1.6 s for vehicles but 135 s for near-static users. The useful window is therefore set by how far a device has moved through the spatially consistent channel, not by how long it took to do so. Vehicles gain least, about 4% against 17 to 28% for the other classes. Each class contains only 7 to 18 test users, so these are single-seed indications, and because step length is fixed in these simulations, the number of samples and the distance covered cannot be separated.

Two checks rule out the most direct alternative explanations. A larger input could help simply because fixed hyperparameters happen to suit it better, so XGBoost and Random Forest were re-tuned separately at every depth, on training users only. Re-tuning changed XGBoost's error by at most 0.44 m and Random Forest's by less than 0.07 m, and the history gain was essentially unchanged (Table X). The derived input channels also matter less than history: replacing them with the five raw channels raised the 1D-CNN's error at $h = 5$ by 0.43 m in a single run, a difference that is suggestive but small beside the gain from history.

With history in place, the choice of model matters far less than the input. At $h = 5$ the six trained models other than k-NN fall within 0.47 m of each other in overall error, with overlapping confidence intervals for every adjacent pair. XGBoost reaches that level in 4 s of training and predicts 1000 positions in a few milliseconds (1.7 ms in one timing and 8.3 ms in another), and

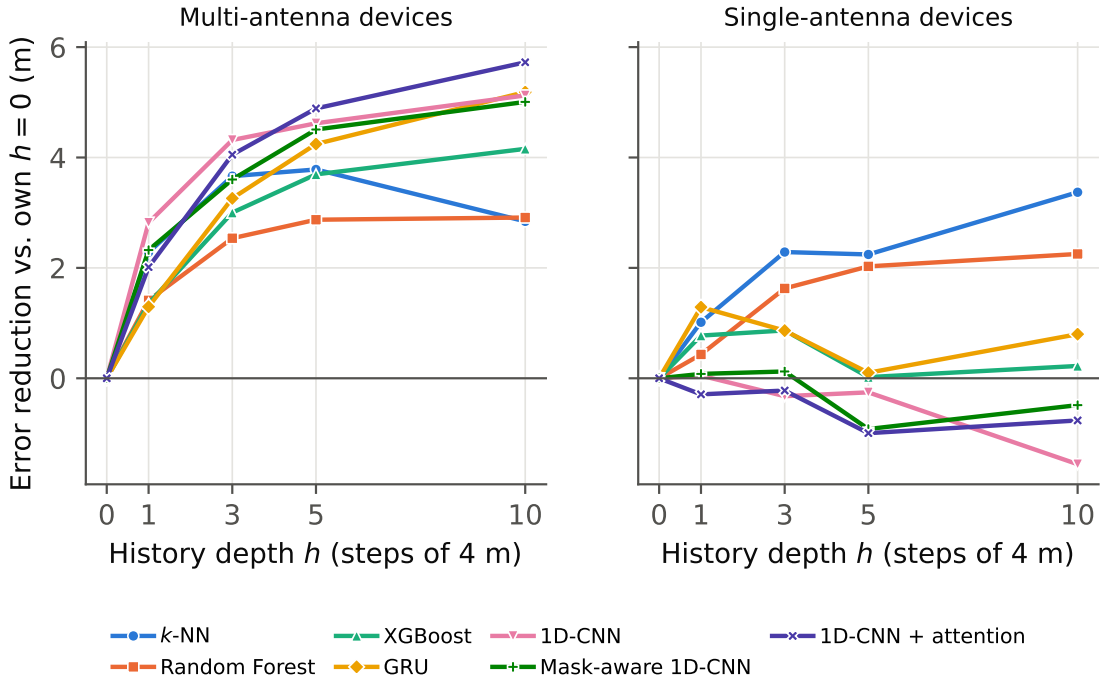


Figure 9: Error reduction against each model's own $h = 0$ input as history depth grows, macro-cell study. Each step of history is $4\,\mathrm{m}$ of travel. Single seed; the single-antenna curves move within that group's seed-to-seed spread, and the five-seed paired values at $h = 5$ are those of Figure 8.

the neural models' errors are, if anything, flattered by their test-based stopping rule. A small boosted-tree ensemble therefore collects the full benefit of history at a fraction of the cost.

Temporal information can also be used after the estimator rather than inside it. A Rauch–Tung–Striebel (RTS) smoother applied to the trained models' predicted tracks lowered their error by a further 2.5 to 4.7%, but with two qualifications: the smoother's noise parameters were tuned on the test users, and its gain comes entirely from the backward pass, which uses future samples. Running only the forward filter raised the 1D-CNN's error by 3.4%. Smoothing is thus a complement to history for offline track reconstruction, whereas history uses past samples only and is available in real time.

All of these gains belong to devices that report an angle. Single-antenna devices are the ones that face the distance-ring ambiguity in its pure form, and they are where history would be expected to matter most.

VIII. Results: Single-Antenna Devices

For single-antenna devices in the macro-cell study, history produced no detectable change in error in any of the seven model families (Figure 8). The seed-paired changes from $h = 0$ to $h = 5$ range from a $1.39\,\mathrm{m}$ increase to a $0.32\,\mathrm{m}$ decrease, and every confidence interval spans zero. The null result is tight where the measurement is precise: XGBoost's change is $0.06 \pm 0.18\,\mathrm{m}$ on a baseline of $35.8\,\mathrm{m}$, and Random Forest's is $0.32 \pm 0.86\,\mathrm{m}$. The same seven models gained 15 to 21% on multi-antenna devices drawn from the same test users. Each split contains between 5 and 10 single-antenna test users, eight on average, which is why their confidence intervals are wider.

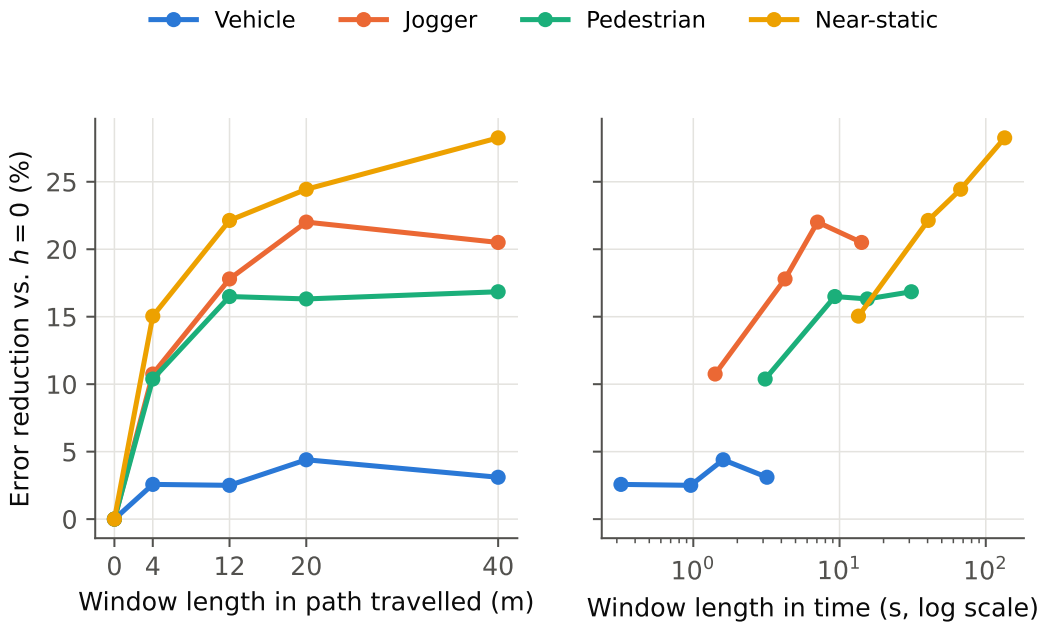


Figure 10: Error reduction against $h = 0$ for each speed class, 1D-CNN, macro-cell study, plotted against the window's length in path travelled (left) and in time (right, logarithmic). Points correspond to $h = 0$, 1, 3, 5 and 10. Single seed, 7 to 18 test users per class.

With 50% of the devices single-antenna, which gives 14 to 22 such test users per split, XGBoost shows a small gain after all, 0.30 ± 0.24 m, and Random Forest none, 0.18 ± 0.24 m. The accurate statement is that history gives these devices no more than about 1% on uniform-NLOS data.

The reason is visible in Section VI: always predicting the mean training position gives 38.5 m for these devices, so a snapshot model at 36.3 m has recovered only about 6% of it, and history leaves that unchanged. Figure 11 shows what the failure looks like for the median single-antenna test user. The user's walk spans 73° of bearing, but the predicted positions, with or without history, span only 17°, at roughly the right distance from the BS (mean predicted range 96 m against 103 m true). This user is illustrative; over all single-antenna devices the predicted range correlates only 0.51 with the true range. The median multi-antenna user's predictions instead follow the shape of the walk, spanning 68° against 65° true. This pattern dominates the tail of the error distribution: in a single-run audit of every test user, all five users with the largest error were single-antenna devices and all five with the smallest were multi-antenna devices.

The 25×25 regression runs, which pool five simulated device profiles, point the same way at the same range, under a different split (Figure 12). With RSS and SINR only and UMi line-of-sight propagation, Random Forest's error rose when three steps of history were added, from 14.5 to 17.1 m under a chronological split and from 14.7 to 18.3 m when the walk was cut into shuffled five-step blocks to prevent any memorization of the path. XGBoost's error fell sharply with history in the same runs, from 29.4 to 13.7 m and from 29.3 to 15.0 m, but from a snapshot model about twice as poor as Random Forest's, to roughly Random Forest's snapshot level; the best history model beat the best snapshot model by only 0.8 m. The result depends on the propagation. Under uniform NLOS, XGBoost's error fell from 33.3 to 24.5 m and Random Forest's from 29.3 to 26.6 m, and with per-cell mixed propagation XGBoost's fell from 29.9 to 20.5 m and Random

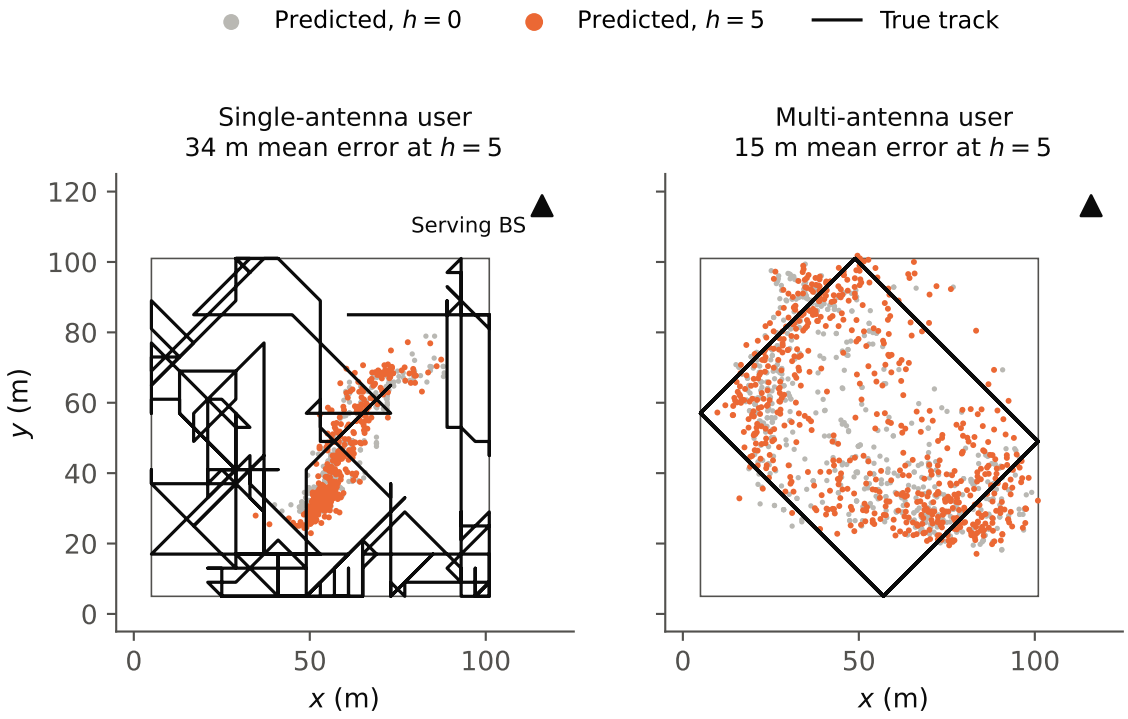


Figure 11: True track and predicted positions of the median single-antenna (left) and median multi-antenna (right) test user, ranked by error at $h = 5$. XGBoost without history (gray) and with five steps of history (orange), macro-cell study, one seed. The box marks the grid; the triangle marks the serving BS.

Forest's from 24.3 to 23.2 m; there the best history model beats the best snapshot model by 4.8 m and 3.9 m respectively, about 16%, although even the best of these errors is 66% and 55% of the 37 m error of always guessing the mean position. Random Forest's loss with history on the line-of-sight data is consistent with a feature-subsampling effect: replacing the stacked absolute lags with differences from the current snapshot removed it in a related configuration with AoA (8.14 against 8.85 m), but the same replacement did not help on the mixed set (11.96 against 11.75 m), and the macro-cell features already include such differences.

A recurrent network changes this picture. Trained on the same data and split, a two-layer GRU over the last $h+1$ reports keeps improving as the window grows to $h = 15$, while the tree ensembles stop gaining after about three steps on line-of-sight data and gain only slowly under NLOS and mixed propagation. At $h = 15$ the GRU has the lowest error in all six combinations of propagation setting and angle: without an angle 12.4, 16.7 and 16.7 m under UMi line of sight, uniform NLOS and mixed propagation, against 13.7, 21.7 and 19.1 m for the best tree model at any depth, and with an angle 6.6, 8.0 and 8.7 m against 8.2, 11.3 and 10.8 m (means of three seeds, which differ by at most 0.4 m). A one-dimensional CNN follows the same course a little above the GRU. In this setting, then, history lowers the absolute error of the best model, not only the error of a weak one. The tree models use parameters tuned at $h = 3$, and the GRU was still improving at its epoch limit, so the size of the gap is indicative rather than final.

How a missing angle is encoded does matter for one kind of model. Feeding single-antenna devices a dummy $0°$ angle makes the derived channels assert a confident bearing, since $\cos 0° = 1$.

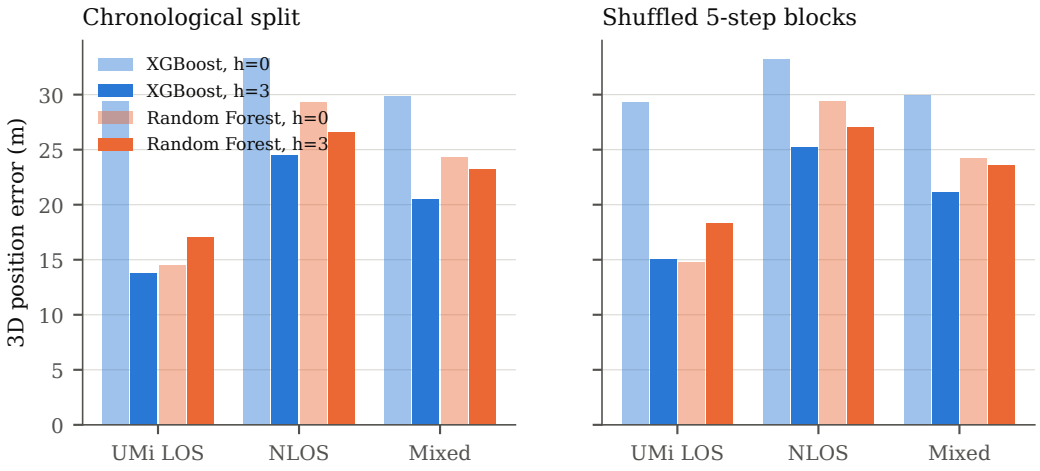


Figure 12: 25×25 regression (five pooled device profiles) with RSS and SINR only: mean position error without history ($h = 0$, light) and with three steps of history ($h = 3$, dark) for XGBoost and Random Forest, under a chronological split (left) and with the walk cut into shuffled five-step blocks (right), for three propagation settings. Single run per bar.

Replacing the angle embeddings with zeros, a point off the unit circle that no real angle can produce, and adding an explicit flag for angle validity lowered the 1D-CNN's single-antenna error from 33.6 to $32.2\,\mathrm{m}$ in a paired single-seed comparison, with a negligible $0.13\,\mathrm{m}$ change for multi-antenna devices. The same encoding did not help k-NN, Random Forest or XGBoost, whose overall error rose slightly in all three cases. A tree can already branch on antenna count and ignore the angle channels when they carry nothing, whereas a convolution applies the same filters to every sample and cannot. The $1.4\,\mathrm{m}$ improvement exceeds the paired uncertainty of the history comparison but lies within the seed-to-seed spread of the single-antenna error, so it is suggestive rather than established.

A. A Range-Dependent Explanation

The contrast with the controlled grid, where RSS and SINR history did help, has a candidate explanation in Equation (7). Without an angle, a history window can only reveal motion through changes in RSS and SINR, and the change produced by one step shrinks with distance from the BS. With the fitted constants, a $2\,\mathrm{m}$ step at $10\,\mathrm{m}$ from the BS changes RSS by about $2.8\,\mathrm{dB}$, whereas a $4\,\mathrm{m}$ step at the macro cell's median range of $95\,\mathrm{m}$ changes it by about $0.6\,\mathrm{dB}$, almost five times less, against shadowing with a spread of $10\,\mathrm{dB}$. An angle measurement weakens with range too, but far less severely relative to its noise: a $4\,\mathrm{m}$ sideways step at $95\,\mathrm{m}$ turns the bearing by about $2.4°$, comparable to the $4°$ angle error. On this account, history helps wherever the step-to-step change in some measured channel stands out from that channel's noise, which at macro range is true of the angle channels and not of RSS.

The account is a hypothesis. The controlled grid and the macro cell differ in other ways as well, most notably in testing seen rather than unseen users, so the range comparison between them cannot by itself isolate the mechanism. It does, however, make three predictions that can be tested within the macro-cell study alone, by feeding the models history for only one group of channels at a time on otherwise identical inputs. These predictions were fixed before that experiment was run. First, for multi-antenna devices, history of the angle channels should recover most of the full

gain, and history of RSS and SINR alone should recover little of it. Second, for single-antenna devices, no subset of history should lower error beyond its confidence interval. Third, whatever gain RSS and SINR history does provide should be largest for devices closest to the BS and should fall with range.

IX. Results: What History Reads

In the macro-cell study, the gain from history is carried almost entirely by the angle channels, and what it mainly reduces is range error. The test fed k-NN, Random Forest and XGBoost history for one group of channels at a time while keeping every other channel at its current value: the RSS group (RSS, SINR and the RSS difference), the angle group (the sines and cosines of both angles and the azimuth difference), or the ray group (the two ray-projection coordinates and their differences). All conditions use the same samples and the same user splits as the full $h = 5$ input, so every comparison is paired, and none of these models selects anything on the test users. The test ran on both the uniform-NLOS and the mixed-propagation realizations, first over five splits for the three models and then over 20 splits for XGBoost and Random Forest. Section IX-A extends it to the four neural models and to what the angle history contains.

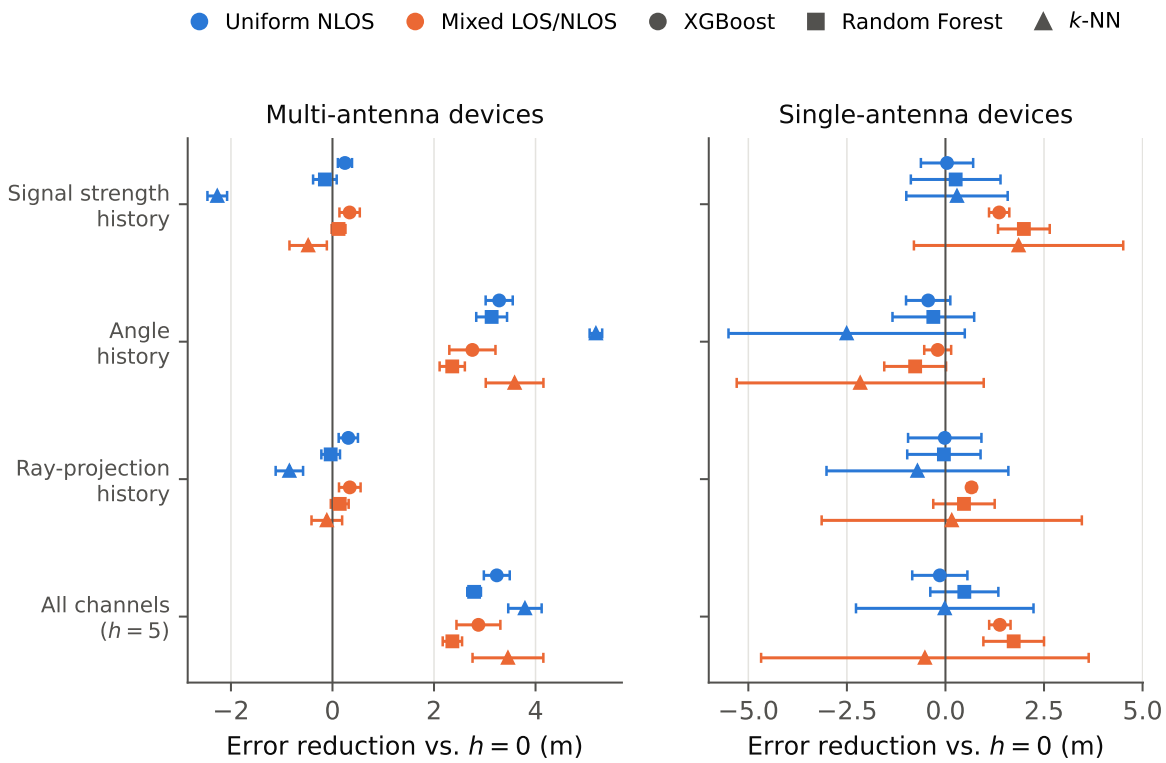


Figure 13: Reduction in mean error against the $h = 0$ input when history is given to one channel group at a time, or to all channels, per device group, model and dataset. Seed-paired means over five user splits with 95% confidence intervals.

The first prediction holds without exception (Figure 13). For multi-antenna devices on the uniform-NLOS data, angle history alone lowered error by 3.28 ± 0.27 m for XGBoost and 3.14 ± 0.30 m for Random Forest, as much as history on every channel (3.24 and 2.79 m). RSS history alone gave 0.25 ± 0.14 m for XGBoost and nothing detectable for Random Forest, and ray history

behaved the same way. For k-NN, angle history alone outperformed the full window (5.19 against 3.79 m), and RSS history made it worse by 2.27 m, consistent with uninformative lags adding noise to its distance metric. The mixed-propagation data repeat the pattern, with angle-only gains of 2.4 to 3.6 m across the three models. Over 20 splits the numbers hold: angle history alone gives 3.16 and 3.06 m for XGBoost and Random Forest, the full window 3.15 and 2.75 m, and RSS history 0.23 and -0.22 m. The four neural networks, given history one channel group at a time by holding the other channels at their current value, show the same pattern over five splits: angle history alone lowers multi-antenna error by 3.0 to 3.4 m, the full window by 2.9 to 3.2 m, and RSS history by under 0.2 m. The angle-channel result therefore covers all seven model families.

What the angle history improves is mainly the estimate of distance (Figure 14). For multi-antenna devices, five steps of history lowered XGBoost's mean radial error from 17.3 to 14.1 m and its tangential error only from 5.0 to 4.6 m; the same holds over 20 splits and for Random Forest and the neural models. Bearing was already well determined by the current angle measurement; the window improves range. Single-antenna devices show no change in either component. Section IX-A asks why a window of angles carries range.

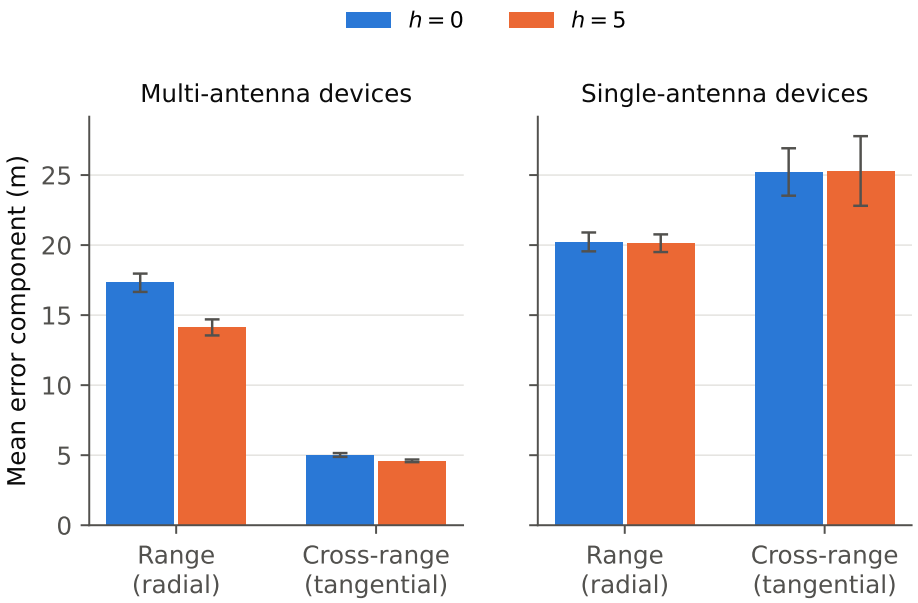


Figure 14: Mean radial (range) and tangential (cross-range) components of the position error without history ($h = 0$) and with five steps ($h = 5$), per device group, XGBoost, macro-cell study. Means over five seeds; bars are 95% confidence intervals.

The test also measures how much history the $h = 0$ input already contains. Removing the first differences from the $h = 0$ input raised multi-antenna error by 0.49 ± 0.14 m for XGBoost and 0.40 ± 0.21 m for Random Forest. The history gains reported in Section VII are therefore measured against a baseline that already holds one step of change, and they understate the full effect of history by roughly 15%.

The second prediction holds on the uniform-NLOS data and fails on the mixed-propagation data. Under uniform NLOS, no channel group lowered single-antenna error for any model over five splits; every confidence interval includes zero. Over 20 splits XGBoost's gains are $0.02 \pm$

0.37 m for RSS history and -0.18 ± 0.34 m for the full window, while Random Forest's full-window gain, 0.42 ± 0.39 m, only just clears zero. Raising the share of single-antenna devices to 50%, which gives 14 to 22 single-antenna test users per split, XGBoost gains 0.30 ± 0.24 m and Random Forest 0.18 ± 0.24 m with the full window. Any gain without an angle is therefore no larger than about 1% of the error. With mixed propagation, RSS history lowered single-antenna error for both tree ensembles, by 1.36 ± 0.26 m for XGBoost and 1.99 ± 0.66 m for Random Forest over five splits and by 1.28 ± 0.29 and 1.41 ± 0.36 m over 20, positive in every one of the 20 splits. The separate five-seed comparison of all seven models on the mixed data confirms this for the two tree ensembles. The neural models share it only weakly: RSS history alone lowers single-antenna error by 0.6 ± 0.4 m for the GRU and 0.9 ± 0.8 m for the attention network, and the full window gives none of the four a gain beyond its interval. Section X examines where this gain arises.

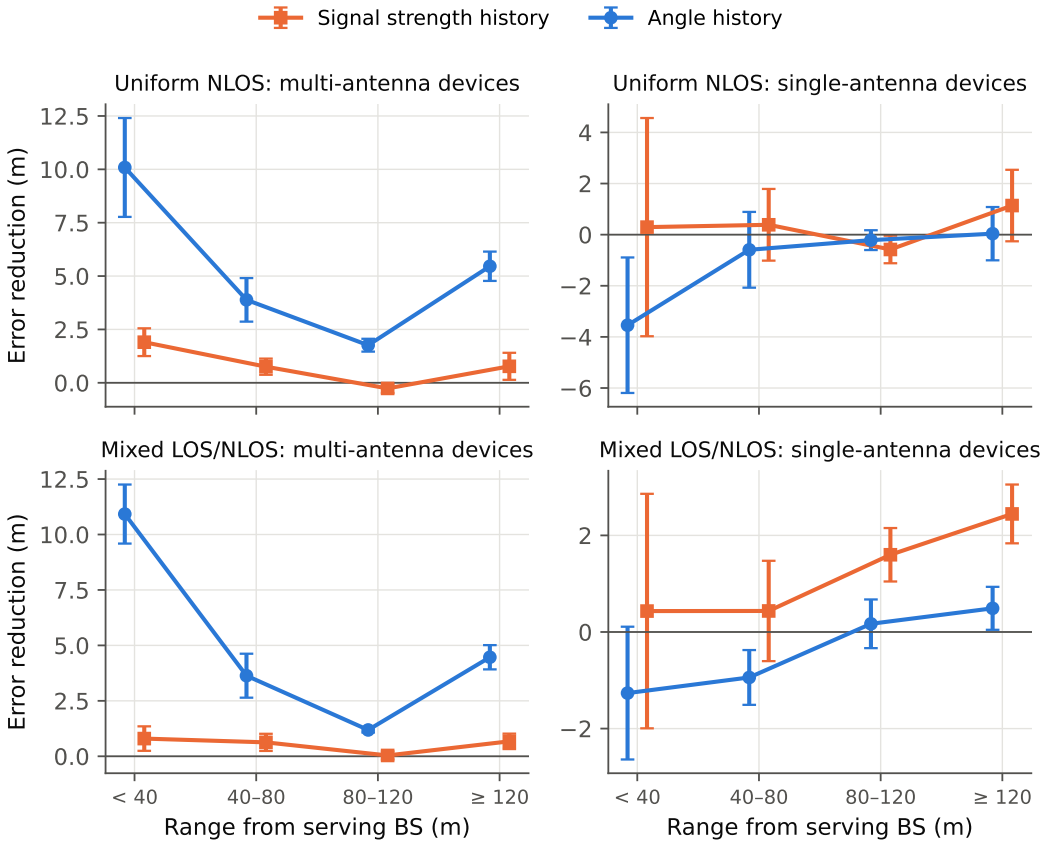


Figure 15: Reduction in mean error against the $h = 0$ input by range from the serving BS, for RSS-only and angle-only history, per device group and dataset, XGBoost. Seed-paired means over five user splits with 95% confidence intervals. The nearest band holds about 160 single-antenna test samples per split. The rise with range of the single-antenna RSS-only gain on the mixed data did not replicate over 20 splits (see text).

The third prediction is not supported, and for XGBoost it is better described as inconclusive (Figure 15). If RSS history were limited by the shrinking gradient of Equation (7), its gain should be largest near the BS and fall with range. For multi-antenna devices on the uniform-NLOS data, XGBoost's RSS-only gain over 20 splits does fall from 0.9 m within 40 m to slightly

below zero at 80 to 120 m, but it rises again to 0.7 m beyond 120 m, and Random Forest's gain has no trend. For single-antenna devices on the mixed data the RSS-only gain is flat across the four range bands at 0.9 to 1.8 m for XGBoost over 20 splits; the rise from 0.4 to 2.4 m seen over five splits did not replicate. The range-dependent account of Section VIII-A therefore does not explain when RSS history helps. Its other half, that the macro-cell gain depends on the angle channels, is the part the test confirms.

The angle-only gain is itself strongly range-dependent. For multi-antenna devices over 20 splits it falls from about 11 m within 40 m of the BS to under 2 m at 80 to 120 m and rises again to 5.4 m beyond 120 m, following the error that is left to remove (22.5, 21.3, 14.4 and 28.2 m without history). In the nearest band it is entirely elevation: 11.0 m of the 11.0 m. This is what the next subsection explains.

A. What the Angle History Contains

Section III-B gives a geometric reason why a window of angles should carry range: for a given sideways movement the bearing turns at the rate $v_\perp/r$. A simpler, statistical reason is also available. Every angle report is the true angle plus an independent 4° error, quantized to 5° steps, so a window offers several noisy views of an angle that has barely changed, and averaging them gives a better estimate. Four tests separate the two, on 20 splits for XGBoost and Random Forest (Figure 16). Replacing the window by the unordered mean of each angle channel gives 3.38 ± 0.19 m and 3.61 ± 0.19 m, as much as the ordered history (3.16 and 3.06 m), so the order of the reports is not needed. Shuffling the order of the lagged reports keeps 2.93 and 2.66 m, a loss of only 0.23 ± 0.05 and 0.40 ± 0.02 m. Elevation history alone gives 2.72 and 2.24 m and azimuth history alone 0.63 and 0.77 m. The base station stands roughly 8.5 m above a device, so elevation encodes distance, and averaging it is a way of improving the range estimate; this is why radial error is what falls.

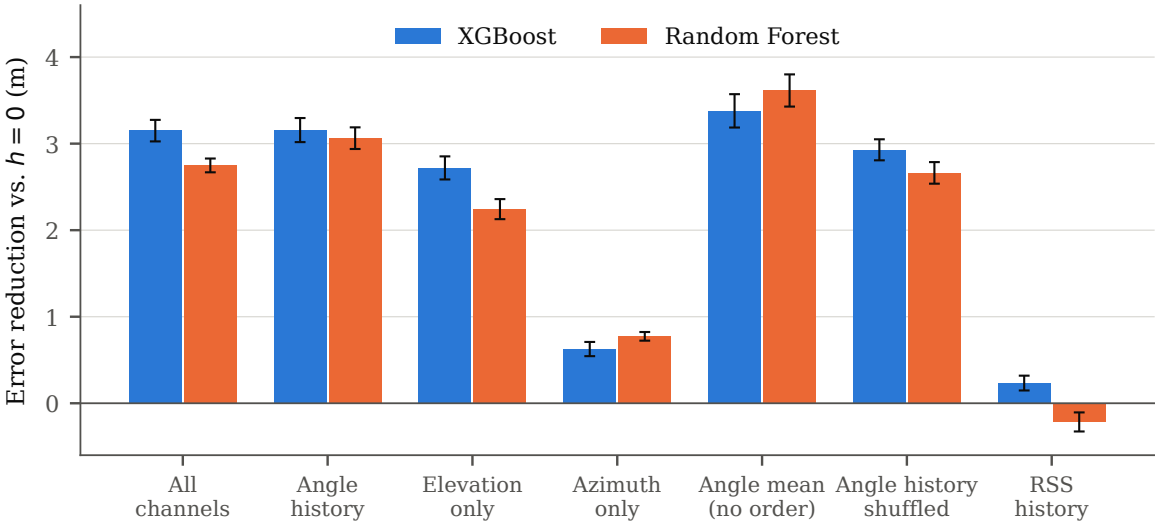


Figure 16: Reduction in mean error against the $h = 0$ input for multi-antenna devices, uniform NLOS, when history is given for different parts of the angle information. "Angle mean" replaces the window by the mean of each angle channel over its six steps; "shuffled" permutes the order of the five earlier reports. Seed-paired means over 20 user splits with 95% Student-t intervals.

The decisive test removes the error. With exact arrival angles the multi-antenna error without history falls to 2.3 m for XGBoost and 1.6 m for Random Forest, and history gives 0.00 ± 0.11 and -1.02 ± 0.03 m: there is nothing left to average. The four neural models behave alike, with errors of 1.8 to 3.2 m and a change of -0.5 to -0.8 m when history is added. Quantization alone, with no noise, leaves the error at 14.2 m and still gives a gain of 1.9 and 2.3 m; noise alone gives 3.1 m (Figure 17a). The history gain of Section VII is thus not a property of bearing geometry. It is a property of averaging imperfect angle reports.

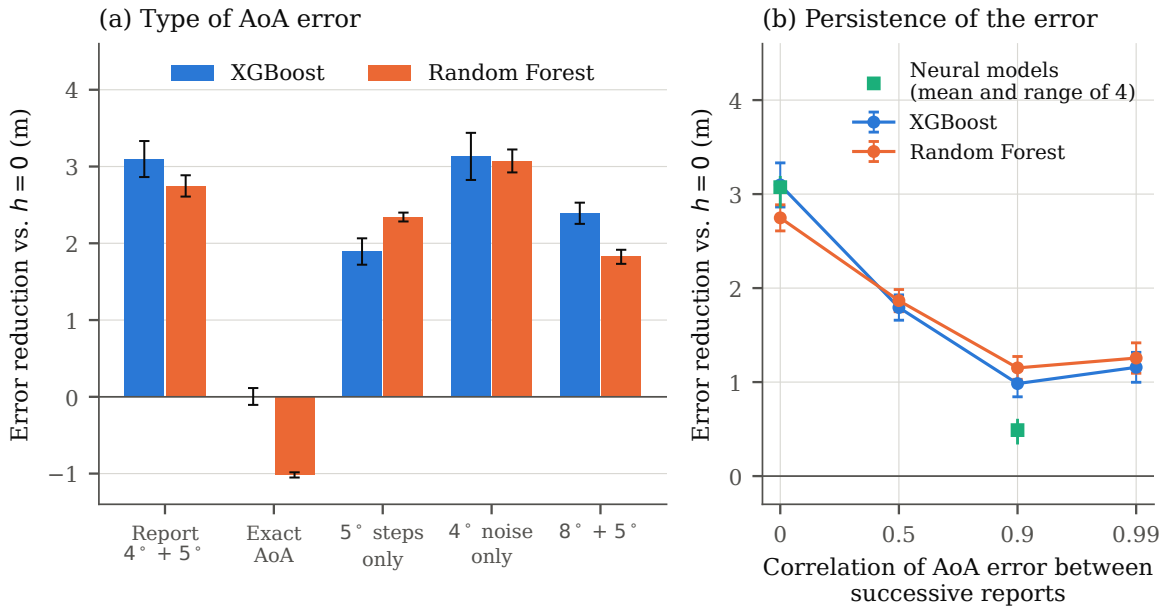


Figure 17: Reduction in mean multi-antenna error from five steps of history, uniform NLOS, for XGBoost and Random Forest, under different angle impairments. (a) Type of error: the report's 4° Gaussian error with 5° quantization, exact angles, quantization only, noise only, and 8° noise with quantization. (b) The 4° error and 5° quantization of the report, with the error correlated between successive reports at the stated lag-1 correlation; squares give the mean and range of the four neural models at the report's setting (correlation 0) and at 0.9. Ten user splits per point except the neural models (five); 95% Student-t intervals.

How much of the gain remains depends on how the error behaves from one report to the next. If the angle error persists across the window, averaging cannot remove it. Making the 4° error an autoregressive sequence along each walk, XGBoost's gain falls from 3.10 ± 0.23 m with independent errors to 1.79 ± 0.13, 0.98 ± 0.14 and 1.16 ± 0.16 m at lag-1 correlations of 0.5, 0.9 and 0.99, and Random Forest's from 2.75 to 1.26 m (Figure 17b). The elevation part disappears first (2.69 to 0.03 m for XGBoost); what remains, about 1 m or 5 to 7% of the error, is in azimuth. It is consistent with the device's motion carrying the true angle across quantization steps, which a window resolves more finely than a single report, but that account was not tested separately. The neural models keep only 0.3 to 0.6 m at a correlation of 0.9, about 2 to 3%. The headline gain of 15 to 21% is therefore specific to angle errors that are independent between reports, which is how the macro-cell simulation draws them; how independent real angle estimates are was not measured here.

The same account explains how the gain depends on window length. Over 10 to 20 splits,

XGBoost's multi-antenna gain from the full window is 1.2, 2.5, 3.2, 3.5 and 3.7 m at 1, 3, 5, 10 and 20 steps (4 to 80 m of travel), so one extra report is already worth 1.2 m and most of the gain is in place by 10 steps. The unordered mean of the angle window is the better input up to 5 steps (3.38 m) and then fails (1.39 m at 20 steps) because over 80 m the true angle has changed and an unweighted mean is biased; the ordered window still gives 3.6 m there, since the model can weight recent reports. At long windows the order of the reports does matter, as a way of discounting stale ones.

X. Results: Propagation, Blockage and Interference

A. *Mixed LOS and NLOS Propagation*

Giving the four regions of the macro cell their own LOS or NLOS propagation leaves the main result intact (Figure 18). About three quarters of test samples now fall in LOS regions. History lowered the overall error of all seven models, by 11 to 15%, with every seed-paired confidence interval clear of zero. For multi-antenna devices the gains are 2.3 to 3.9 m, as on the uniform-NLOS data, and the six trained models other than k-NN again lie within 0.4 m of each other. The single-antenna result differs: both tree ensembles now gain about 1.6 m for single-antenna devices over five splits and 1.1 to 1.3 m over 20, while the neural models and k-NN do not gain beyond their intervals.

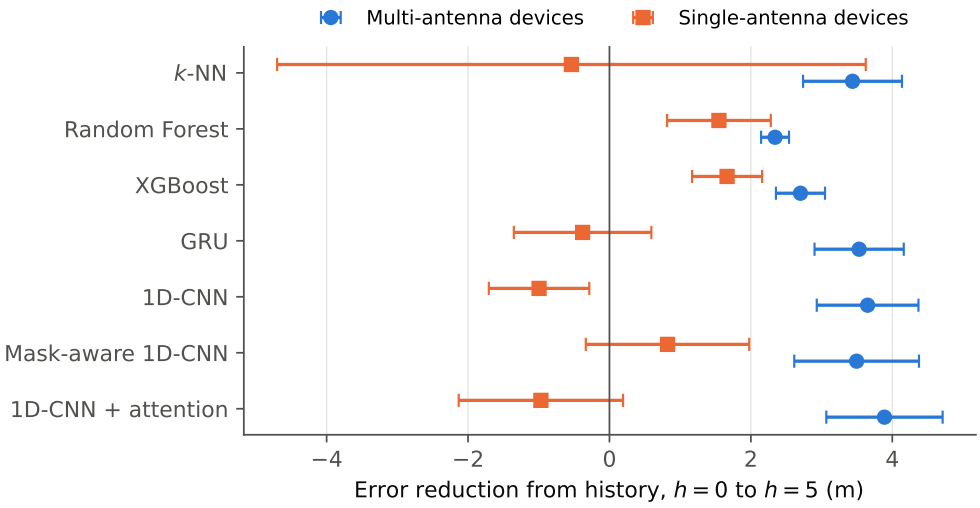


Figure 18: Reduction in mean error from adding history ($h = 0$ to $h = 5$), per model and device group, mixed-propagation realization of the macro-cell study. Seed-paired means over five user splits with 95% confidence intervals.

Signal strength is more informative about range in the LOS regions. A path-loss fit to the LOS samples gives $\gamma = 2.58$ with a shadowing spread of 7.7 dB, and range explains 26% of the variance in RSS. In the NLOS regions the fit gives $\gamma = 3.60$, a spread of 11.9 dB, and only 7% of the variance explained.

History helps more in the regions that carry LOS propagation, but this is a property of the regions and not of line-of-sight propagation (Figure 19). In a single XGBoost run, history lowered error by 15.9% in the two LOS regions and by 8.7% in the two NLOS regions. The uniform-NLOS realization of the same walks contains no LOS at all, and the same regions give 16.0% and 4.8%. Over 20 splits, for multi-antenna devices, XGBoost's gain in the LOS regions is

3.56 ± 0.17 m on uniform NLOS and 3.10 ± 0.19 m with real LOS, and in the NLOS regions 1.75 ± 0.16 and 1.55 ± 0.13 m. The gap between regions is 1.81 ± 0.27 m without any LOS and 1.54 ± 0.24 m with it, a difference of -0.27 ± 0.25 m, and Random Forest gives -0.01 ± 0.18 m. Making the propagation real did not widen the gap. The regions differ in range and in the error left to remove, since the gain is largest in the nearest and farthest range bands (Section IX), and these are where the LOS regions lie. Earlier readings of the zone comparison as support for a LOS effect should therefore be discarded.

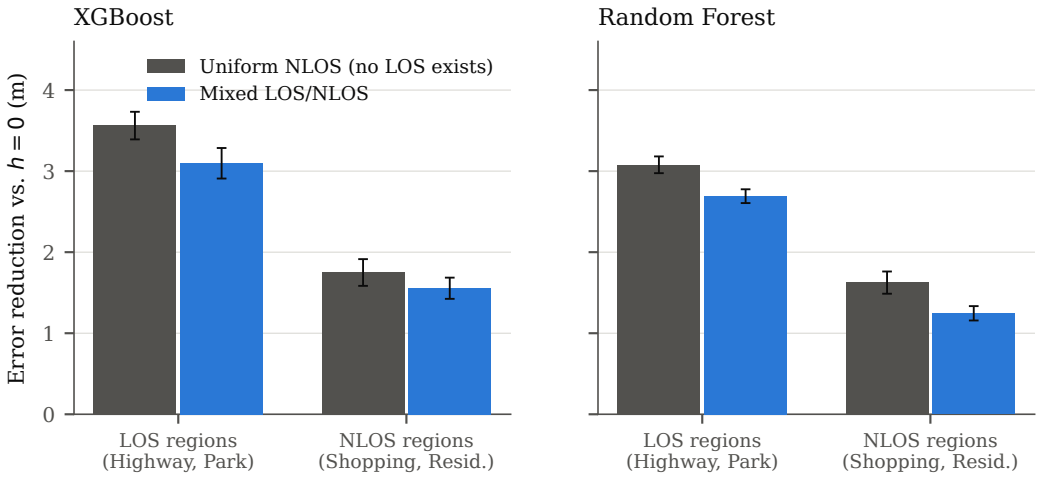


Figure 19: Reduction in mean error against the $h = 0$ input for multi-antenna devices, full history, in the two regions that carry LOS propagation in the mixed realization (Highway, Park) and the two that carry NLOS propagation (Shopping, Residential), and in the same regions of the uniform-NLOS realization, which contains no LOS. Seed-paired means over 20 user splits with 95% Student-t intervals.

The single-antenna gain from RSS history on these data is not explained by the cleaner LOS path-loss relation: over five splits XGBoost's RSS-only gain is 1.28 ± 0.42 m for samples in LOS regions and 1.53 ± 0.87 m for samples in NLOS regions. What it does depend on is the propagation state itself. A single-antenna snapshot cannot tell LOS from NLOS, which differ by about 17 dB in RSS, and adding the true LOS flag to the snapshot as an oracle lowers error by 3.27 ± 0.49 m for XGBoost and 3.04 ± 0.44 m for Random Forest over 20 splits. RSS history recovers part of this, 1.28 ± 0.29 and 1.41 ± 0.36 m, and given on top of the oracle flag it still gains 1.01 ± 0.27 and 0.77 ± 0.25 m. That remaining gain is concentrated in the 14% of windows that span a region crossing, where it is 2.6 and 3.9 m, against 0.8 and 0.3 m for windows entirely in LOS regions and 0.6 and 0.0 m for windows entirely in NLOS regions. A consistent reading is that a window both helps to infer the propagation state and, at a crossing, marks a place on the map where the signal strength steps; the models were not examined to confirm either mechanism. The gain is positive in every one of the 20 splits for both ensembles, and with half of the devices single-antenna, which gives 14 to 22 test users per split, it is 1.20 ± 0.15 and 1.42 ± 0.15 m.

B. A Blocked Line of Sight

The blocked line-of-sight scenario was designed to test whether history becomes more valuable as a blockage lengthens, up to the point where the heading observed before the blockage stops being informative, after which its value should decay. Over 20 splits, with the two device groups

analysed separately, history lowered the error of multi-antenna devices by 6.29 ± 0.18 m (36%) inside the blocked stretch and by 6.66 ± 0.15 m (39%) outside it for XGBoost, and by 6.13 ± 0.18 and 6.16 ± 0.12 m for Random Forest. The blocked stretch is therefore not special. The reduction does not rise and fall with the width of the blocked strip (Figure 20): it lies between 5.2 and 6.1 m from 5 to 120 m and reaches 7.7 m at the widest strip, 150 m, so the predicted dependence on blockage length does not appear. Single-antenna devices gain nothing detectable, 0.47 ± 0.54 m inside the stretch for XGBoost.

The size of the gain has the same source as in the macro cell. The serving BS stands 500 m from the corridor, so an azimuth step of 5° spans tens of meters across it, and the unordered mean of the angle window recovers all of the gain (6.51 ± 0.18 m inside the stretch). The scenario shows that history keeps its value through a blockage of any length tested, but it does not isolate an effect of blockage, because the same gain appears outside the blocked stretch.

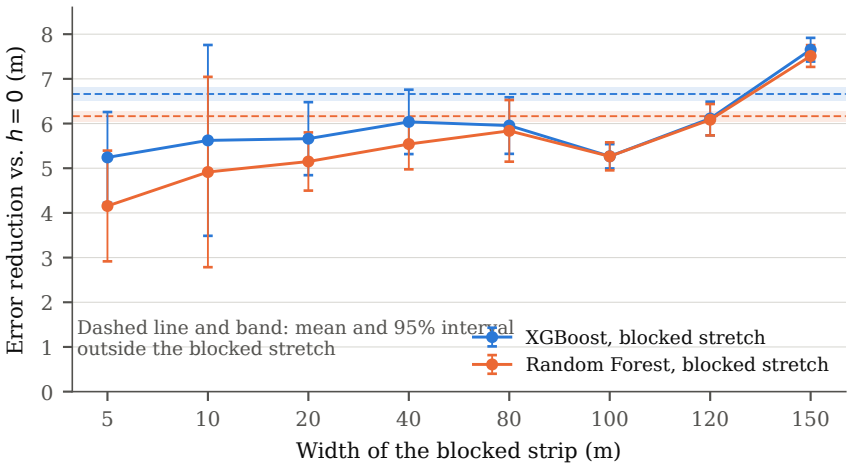


Figure 20: Reduction in mean error from adding history ($h = 0$ to $h = 5$) for multi-antenna devices inside the blocked stretch, by width of the blocked strip. Dashed lines and bands give the mean and 95% interval outside the blocked stretch. Seed-paired means over 20 user splits with 95% Student-t intervals.

C. Without the Interfering Base Stations

Both macro-cell realizations contain two interfering BSs, and the cell is fully interference-limited: median SINR is -3 dB, against a median interference-free signal-to-noise ratio of about 32 to 46 dB. The interferers enter only through SINR, so switching them off is the same simulation with SINR replaced by that ratio, which follows exactly from the stored RSS; no new channel simulation is needed. Table V gives the result for twenty user splits. For multi-antenna devices the interferers are worth between 0.6 and 1.0 m without history, and the history gain is slightly larger without them, by 0.3 to 0.4 m; angle history still carries all of it, 3.50 ± 0.18 m of the 3.44 m for XGBoost on uniform NLOS. For single-antenna devices removing them changes the error by under 1 m on uniform NLOS and costs 2.9 m on mixed data, where the history gain of the previous subsection survives unchanged (1.04 ± 0.23 against 1.14 ± 0.29 m for XGBoost). The conclusions of this report therefore do not depend on the interfering stations. Placing them elsewhere was not tried.

Table V: Multi-antenna error without history and the gain from five steps of history, with and without the two interfering base stations, XGBoost and Random Forest, twenty user splits. Gains are seed-paired means with 95% Student-t intervals.

		Error at $h = 0$		Gain from history	
Dataset	Model	with	without	with	without
Uniform NLOS	XGBoost	18.74	19.30	3.15 ± 0.12	3.44 ± 0.17
Uniform NLOS	Random Forest	18.76	19.34	2.75 ± 0.08	3.15 ± 0.11
Mixed	XGBoost	17.48	18.46	2.75 ± 0.15	3.15 ± 0.17
Mixed	Random Forest	17.15	18.14	2.36 ± 0.08	2.80 ± 0.09

XI. Discussion

a) What history reads.

The motivating account, that the rate at which the bearing turns reveals range, is not what the data show. The gain of the multi-antenna devices comes from the angle channels in every model family, but an unordered mean of the window reproduces it, elevation carries most of it, and it vanishes when the angles are exact. History is useful here because it lets an estimator average several noisy, coarsely quantized angle reports; since elevation encodes distance, the averaging appears as lower range error. This is a smaller claim than the one the study set out to test, and a more reliable one, because it was tested directly on seven model families. On the controlled grid the account is incomplete: sequence models keep gaining from longer windows where tree ensembles, even when given the window mean, do not, and shuffling the older reports, with the current one kept in place, removes most of the gain from long windows, so a part of the grid gain depends on the order of the reports. Whether that part exists at macro range, where one report carries much less, is not known.

b) How far the result extends.

The gain is as large as it is because the simulated angle error is drawn independently for every report. With an error that persists across the window it falls from about 17% to 6% for the tree ensembles and 2 to 3% for the neural models. A real angle estimator's errors are neither fully independent nor fully persistent, and measured traces are needed to say where between the two it lies. Until then the report's headline should be read as the upper end of a range whose lower end is between a sixth and a third of it.

c) Devices without an angle.

On uniform NLOS a single-antenna device is barely localized at all: its models are only 5 to 6% better than guessing the cell centre, with or without history. Signal strength from one base station, under about $10\,\mathrm{dB}$ of shadowing, carries little positional information over a $96\,\mathrm{m}$ cell. Under mixed propagation the tree ensembles gain about $1.2\,\mathrm{m}$, and an oracle experiment links that to the propagation state, which a single-antenna snapshot cannot tell apart; the neural models share it only weakly. The earlier hypothesis that the benefit of RSS history shrinks with range was not supported.

d) The two regimes.

The controlled grid and the macro cell differ in scale, in the information one RSS reading carries, and in whether the test users were seen in training. The grid result, that history helps without an angle, and the macro-cell result, that it does not, are consistent with each other but do not isolate the cause, and the macro-cell data did not support the one candidate cause that was tested directly. The 25×25 regression runs without an angle are mixed rather than negative

and depend on the propagation: on line-of-sight data Random Forest degrades with history while XGBoost halves its error, to about the level of Random Forest's snapshot model; under uniform NLOS and mixed propagation both gain, XGBoost by 8.7 and 9.4 m and Random Forest by 2.7 and 1.1 m, from no-history errors of 33.3 and 29.9 m (XGBoost), against 37 m for always guessing the mean position.

e) Practical case.

With history in place the choice of model matters far less than the input, and a boosted-tree ensemble that trains in seconds collects the full gain. The conditions that this report found not to matter, line-of-sight propagation, a blocked line of sight, and the two interfering base stations, are the ones a deployment is most likely to vary.

XII. Limitations

All results come from simulation. Spatial consistency, the property that gives a measurement sequence its information, is a feature of the channel model here; real channels exhibit it, but also add non-stationarity, hardware impairments and calibration drift that the simulations omit. The trajectories are likewise synthetic: they are generated on a grid graph rather than drawn from measured traces, and lack the road geometry and stop-and-go behavior of real movement.

The two device groups of the macro-cell study share one simulated channel, since every UE is simulated with the same omnidirectional array and the absence of an angle is imposed when features are built. The contrast between the groups is therefore a contrast in available measurements rather than in physical hardware. The $85/15$ population mix is an assumption, which is why every result is reported per group rather than for the blend. Each test split of 47 users contains only 3 to 11 single-antenna devices, so their errors carry much wider seed-to-seed uncertainty than those of multi-antenna devices, and any result resting on the nearest range band rests on a few hundred single-antenna samples at most.

The angle-channel result rests on 20 user splits for XGBoost and Random Forest and on five for k-NN and the four neural models. The central result also depends on the simulated angle error: every report carries an independent 4° Gaussian error, quantized to 5° steps, and Section IX-A shows that the gain falls from about 17% to 2 to 6% when that error persists across reports. How correlated the errors of a real angle estimator are was not measured, and the correlated case was run for the trees at four correlations and for the neural models at one. The single-antenna gain that the tree ensembles obtain from RSS history under mixed propagation is linked to the propagation state by an oracle experiment, not by an examination of what the models infer, and five splits cannot settle whether the neural models share it. The interfering base stations were switched off but not moved.

Because every step of a walk advances exactly one grid cell, history depth measures both the number of samples and the distance travelled, and the two cannot be separated in these data; doing so requires simulations with variable step length or variable sampling rate. The neural models chose their stopping epoch on the test users, which flatters their absolute errors and any comparison with the tree and instance-based models, though not the comparisons between depths of one model. Finally, the controlled-grid tree results are single runs, and repeated runs of the same configuration differ by two to three percentage points, so only their larger effects are interpreted; the CNN and GRU curves rest on three seeds and the shuffled-window GRU on two. The neural stopping remark above concerns the macro-cell study; on the controlled grid and in the 25×25 runs the epoch is chosen on a validation block taken from the training data.

The absolute controlled-grid numbers depend strongly on the propagation setting. The grid was regenerated after two defects of the generator were found, a carrier frequency that was never

applied and Voronoi cells that never changed the propagation scenario, and it was simulated under three settings: rural-macro line of sight, a per-cell mix of urban-micro line of sight and NLOS, and uniform NLOS. With the BS beyond the grid corner, XGBoost with three steps of history and no angle reaches 48.6, 57.3 and 25.8% exact-cell accuracy under the three settings. The direction of every history effect is the same in all three, but no single absolute level should be read as the accuracy of the method. The earlier controlled-grid simulations, which gave higher accuracies, cannot be regenerated from the repository and are not used.

The sequence finding of the controlled grid is limited in three ways. First, it was tested on the grid and in the 25×25 regression runs but not in the macro-cell study; at macro range the gain is still explained by the averaging of angle reports, and whether order carries information there is open. Second, the order test keeps the current report in place and shuffles the older ones; it shows that their order matters but not which aspect of it, such as the direction of movement or the rate of change, and it rests on two seeds and one architecture of each kind. Third, the window-mean control rules out one explanation for the plateau of the tree ensembles, a missing average, but does not identify the cause; overfitting to many correlated lag columns remains possible. Finally, the studies do not share one angle-error model: the controlled grid uses an error that grows as the SINR falls, while the macro-cell study and the 25×25 runs cited here use a fixed 4° error, and the 25×25 results with an angle change markedly between the two models.

The 25×25 regression runs pool five device profiles in one model and are single runs per propagation setting. The profiles are not equally easy to locate: in the line-of-sight set one profile has a mean error of 13.7 m with history and an angle, about twice the 6.4 to 7.4 m of the others, so the pooled error hides differences between devices.

XIII. Conclusion

In a simulated single-BS macro cell, five steps of transition history lower the position error of devices that report an angle of arrival by 15 to 21% in all seven model families tested, and the choice of model matters far less than the input. The gain comes from the angle channels and is the averaging of noisy angle reports, chiefly elevation, which shows up as lower range error; it vanishes with exact angles and shrinks to 2 to 6% when the angle error persists across reports. Devices without an angle gain no more than about 1% on uniform NLOS, where they are barely better than a constant guess, and about 1.2 m under mixed propagation for tree ensembles. Line-of-sight propagation, interfering base stations and a blocked line of sight do not change the multi-antenna result. On the controlled grid, history helps under every propagation setting and placement tested, tree ensembles stop gaining at about three steps, and a GRU and a one-dimensional CNN keep improving up to fifteen; shuffling the older reports inside the window, with the current report kept in place, removes most of their gain from long windows, which suggests that the order of the reports, and not only their mean, carries information about position there.

Three follow-ups would settle what remains. Measured angle-of-arrival traces would show how correlated real angle errors are, which sets where the gain lies between the independent and persistent cases. Simulations with variable step length or sampling rate would separate the number of reports from the distance they cover. A direct examination of what the tree ensembles infer from RSS history under mixed propagation would test the account offered for the single-antenna gain.

APPENDIX A
CODE AND DATA AVAILABILITY

The simulation configurations, generation scripts, feature pipeline, model definitions and analysis scripts are available in the project repository (https://github.com/slash827/CSI-Location). Simulated datasets are regenerated from the configurations rather than distributed. Every number in this report is drawn from machine-written summary files kept in the repository alongside an index of the experiment that produced each one. The global random seed is 42; the five-seed repeats use seeds 42, 1, 7, 13 and 99.

APPENDIX B
MODEL HYPERPARAMETERS

Table VI: Macro-cell model configurations. Hyperparameters are identical at every history depth.

Model	Configuration
k-NN	$k = 5$ neighbors, Euclidean distance on standardized features
Random Forest	150 trees, maximum depth 16, minimum leaf size 5, $\sqrt{d}$ features per split
XGBoost	50 trees, maximum depth 5, learning rate 0.15, row and column subsampling 0.8, one output per tree
1D-CNN	Three convolutional layers (32, 64, 128 filters, kernel 3) with batch normalization and dropout 0.35, global average pooling, a shared 128-unit trunk with the static device features, and position, speed and uncertainty heads; $66{,}500$ parameters
Mask-aware 1D-CNN	As the 1D-CNN, with an additional AoA-validity input channel; $66{,}596$ parameters
1D-CNN + attention	As the 1D-CNN, followed by a four-head temporal self-attention block with learned positional embeddings; $199{,}748$ parameters
GRU	Two recurrent layers of 128 units, dropout 0.30, a 256-unit shared trunk and the same three heads; $254{,}468$ parameters
Neural training	AdamW, learning rate 3×10^{-4}, weight decay 10^{-2}, batch size 512, Huber loss with weights 1.0, 0.2 and 0.05 on position, speed and uncertainty, at most 60 epochs, early stopping with patience 8

APPENDIX C
SUPPLEMENTARY RESULTS

Table VII: Effect of history ($h = 0$ to $h = 5$) per model and device group, macro-cell study. Errors are means over five seeds; gains are seed-paired, with 95% confidence intervals. Training time and time to predict 1000 positions come from a single run.

	Multi-antenna devices		Single-antenna devices		Cost	
Model	Error, $h = 0$	Gain	Error, $h = 0$	Gain	Training	ms per 1000
k-NN	23.4 m	3.77 ± 0.33 m	46.2 m	-0.11 ± 2.32 m	1 s	577
Random Forest	18.8 m	2.75 ± 0.13 m	36.6 m	0.32 ± 0.86 m	34 s	102
XGBoost	18.8 m	3.16 ± 0.34 m	35.8 m	0.06 ± 0.18 m	4 s	1.7
GRU	19.3 m	3.54 ± 0.66 m	36.7 m	-0.26 ± 0.76 m	166 s	31
1D-CNN	19.5 m	3.73 ± 0.85 m	36.0 m	-1.16 ± 1.82 m	197 s	14
Mask-aware 1D-CNN	19.3 m	3.47 ± 0.75 m	34.8 m	0.25 ± 0.82 m	182 s	18
1D-CNN + attention	19.5 m	4.06 ± 0.74 m	35.9 m	-1.39 ± 3.28 m	212 s	23

Table VIII: Macro-cell study: overall mean error (m) by history depth for every model, single seed. These are the values behind Figure 9, pooled over both device groups.

Model	$h = 0$	$h = 1$	$h = 3$	$h = 5$	$h = 10$
k-NN	27.44	25.43	24.07	23.98	24.49
Random Forest	22.11	20.91	19.77	19.42	19.34
XGBoost	22.35	21.12	19.80	19.43	19.03
GRU	22.72	21.43	19.97	19.36	18.47
1D-CNN	22.85	20.62	19.52	19.27	19.15
Mask-aware 1D-CNN	22.10	20.26	19.24	18.75	18.26
1D-CNN + attention	22.54	21.02	19.40	18.90	18.19

Table IX: Macro-cell study: overall mean error (m) over five seeds, with 95% confidence interval and full range across seeds, and the seed-paired overall gain from $h = 0$ to $h = 5$.

Model	$h = 0$	Spread, $h = 0$	$h = 5$	Spread, $h = 5$	Paired gain
k-NN	27.07 ± 0.64	1.85	23.86 ± 0.85	2.53	3.22 ± 0.27
Random Forest	21.66 ± 0.46	1.24	19.28 ± 0.39	1.19	2.38 ± 0.20
XGBoost	21.51 ± 0.50	1.35	18.87 ± 0.46	1.12	2.65 ± 0.29
GRU	22.03 ± 0.56	1.27	19.09 ± 0.49	1.36	2.95 ± 0.49
1D-CNN	22.10 ± 0.62	1.46	19.12 ± 0.52	1.31	2.98 ± 0.63
Mask-aware 1D-CNN	21.74 ± 0.59	1.56	18.81 ± 0.52	1.46	2.93 ± 0.58
1D-CNN + attention	22.08 ± 0.61	1.61	18.84 ± 0.44	1.30	3.24 ± 0.63

Table X: Macro-cell study: overall mean error (m) with fixed hyperparameters and with hyperparameters re-tuned separately at each depth on training users only, single seed.

	XGBoost		Random Forest	
Depth	Fixed	Re-tuned	Fixed	Re-tuned
$h = 0$	22.35	21.91	22.11	22.05
$h = 1$	21.12	21.20	20.91	20.84
$h = 3$	19.80	19.66	19.77	19.81
$h = 5$	19.43	19.07	19.42	19.40
$h = 10$	19.03	18.97	19.34	19.30